Self-Defeating Public Investment Cuts: Evidence from Poland under EU Fiscal Surveillance

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Abstract

EU fiscal surveillance evaluates national budgets through debt projections and net expenditure ceilings. This leaves public investment particularly vulnerable to postponement, because reducing or delaying investment is often politically preferred to raising taxes or to cutting transfers, wages, and entitlements. This paper analyses how estimated public investment responses influence debt implications for Polish investment policy by evaluating fiscal transmission through individual transmission channels: investment import content, household liquidity position, public debt, and real PPP income level. Annual local projections for public-investment shocks are estimated within an EU27 panel framework. The Polish scenario keeps a separate evaluation for each of three channels, investment import content, household liquidity position, and public debt, when the eighth-year output interaction satisfies the stated retention criterion; real PPP income level is not retained.

These estimated responses form the basis of a three year Polish scenario, in which annual public investment is either increased or reduced by one percentage point of GDP from 2028 to 2030. By the eighth year, cumulative public-investment multipliers reach 2.87 in the EU27 benchmark, ranging from 2.37 to 2.91 across the retained Polish channel evaluations, with an equal weight average of 2.59. By 2036, a central paradox becomes evident: public-investment expansion reduces the debt-to-GDP ratio relative to the Commission baseline, while an equivalent public-investment cut, intended as consolidation, increases it. Across the retained Polish channel evaluations, the cut scenario raises the 2036 debt-to-GDP ratio by 3.7 to 7.3 percentage points under the institutional debt equation, and by 1.7 to 3.5 percentage points under the direct debt-to-GDP local projection path. Together, these two results reveal a self-defeating mechanism in which the discretionary primary-balance gain from cutting investment can be outweighed by persistent output losses, related cyclical primary-balance deterioration due to automatic stabilisers, and the denominator effect on the debt ratio.

1 Introduction

1.1 EU fiscal surveillance and debt sustainability analysis

EU fiscal surveillance evaluates member states' fiscal plans by applying legal rules, numerical indicators, medium-term adjustment trajectories, and model-based assessments within the Stability and Growth Pact and the European Semester. This framework sets the institutional context in which the European Commission and the Council measure fiscal efforts and scrutinise national budgetary decisions (Schmidt, 2015; Van der Veer, 2021; European Commission, 2026). Recent reforms to the EU fiscal framework strengthen the role of debt sustainability analysis in providing prior guidance, assessing medium-term budgetary plans, facilitating bilateral discussions with member states, and defining corrective trajectories where fiscal risks are considered significant (European Commission, 2026; Heimberger et al., 2024).

The institutional significance of the Commission's framework arises from its surveillance role. Structural balances, potential output, output gaps, projected debt paths, interest rate assumptions, growth forecasts, and fiscal multipliers are all model-dependent. When integrated into the surveillance model, adjustments to output gap estimates can alter the calculated structural balance and measured fiscal effort even without contemporaneous changes in taxes, expenditures, or headline budget balances. Heimberger, Huber and Kapeller (2020) show this within the Commission's potential output model: assumptions about production functions, labour input trends, and capacity utilisation affect output gap estimates, which in turn move structural deficit estimates and available fiscal space. Methodological decisions therefore shape surveillance assessments by shaping the evaluation of national fiscal plans.

The Debt Sustainability Monitor 2025 sets out the Commission's medium-term projection methodology. It combines Commission forecasts with assumptions on potential growth, output gap convergence, structural primary balances, inflation, interest payments, ageing-related expenditures, stock-flow adjustments, deterministic stress scenarios, and stochastic risk analyses (European Commission, 2026). The empirical approach in this paper does not estimate an output gap equation directly but instead estimates real GDP responses explicitly, treating the Commission's output gap convergence assumption solely as a component within the evaluated baseline projection environment. Macroeconomic feedback from fiscal interventions enters through GDP, cyclical budget elements, subsequent primary balance trajectories, and the debt dynamics driven by the interest-growth differential. Within the reformed framework, these projections support prior fiscal guidance, assessments of medium-term structural fiscal plans, Council-approved net expenditure ceilings, annual progress evaluations, and corrective pathways associated with the excessive deficit procedure.

The excessive deficit procedure carries institutional authority because non-compliance can trigger Council recommendations, an assessment of effective action, a revised fiscal adjustment trajectory, closer monitoring, and further escalation under the corrective arm. Euro area member states are also subject to legal provisions for deposits or financial penalties. Poland is outside the euro area, so direct financial sanctions do not apply immediately. Compliance pressure on Poland therefore runs through Council-approved adjustment requirements, annual surveillance, and procedural determinations about effective action. A conditional legal channel remains: macroeconomic conditionality may affect access to cohesion policy funding where the relevant economic governance criteria are met (Sacher, 2019; European Commission, 2026; Council of the European Union, 2024, 2025).

The baseline is the Commission's Debt Sustainability Monitor projection, produced within the EU fiscal surveillance process. This paper adopts that baseline as an externally given reference path and does not re-estimate it. The path incorporates the Commission's forecast round and assumptions on potential growth, output gap closure, inflation, interest expenditure, ageing-related costs, stock flow adjustments, structural balances, primary balances, and the debt dynamics equation. Under these assumptions, Poland's general government gross debt ratio rises from 55.1 percent of GDP in 2024 to 106.8 percent in 2036, surpassing the 60 percent Treaty reference value. The same endpoint corresponds to a structural balance near -10 percent of GDP and a primary balance around -3.9 percent of GDP.

The implausibility is arithmetic before it is behavioural, and it can be read directly off the Commission's own inputs. The baseline's implied nominal interest rate on government debt rises from about 4.8 percent in 2025 to about 6.2 percent in 2036, and the implied real rate rises from roughly 1.2 percent to roughly 3.4 percent over the same span. The interest-growth differential, favourable in the early years, changes sign in the course of the projection: nominal growth exceeds the implied interest rate until 2029 but falls short of it from 2030 onward, so the snowball term that initially lowers the ratio turns into one that raises it. By 2036 the differential alone adds about 0.9 percentage point to the debt ratio each year before any primary deficit is counted. The decomposition of the change in the debt ratio into a snowball term, the primary balance, and the stock-flow adjustment, together with the condition that the ratio stabilises only when the primary balance roughly matches the differential times the lagged debt ratio, makes the gap explicit (Checherita-Westphal and Domingues Semeano, 2020). Evaluating that condition at the 2036 debt ratio of 106.8 percent of GDP and the baseline's own 2036 differential, stabilising the ratio would require a primary surplus of about one percent of GDP, whereas the baseline assumes a primary deficit near -3.9 percent.

The distance between the assumed balance and the debt-stabilising balance is therefore close to five percentage points of GDP, and it must be closed merely to stop the ratio from rising, before any reduction. The baseline closes none of it: the primary deficit stays inside a narrow band of roughly -3.6 to -4.3 percent of GDP across the whole horizon, and a structural balance near -10 percent of GDP at the endpoint is not a level any government holds as a stable stance.

Reading the result through the actors that finance and govern the path points the same way: investors, EU institutions, and domestic policymakers would all have to be assumed not to react to a debt ratio rising past the 60 percent Treaty reference value while the excessive deficit procedure stays active. The path is therefore a Commission surveillance construct: a no-additional-measures projection that holds corrective measures outside the baseline so that fiscal risks can be measured against a common reference path. Poland's domestic fiscal regime and medium term fiscal structural plan are consistent with this reading, but the arithmetic is the load-bearing point: the baseline endpoint is a surveillance counterfactual under strained debt dynamics, not a fiscal path that the Polish government could select and sustain (Blanchard, 2019; Huidrom, Kose, Lim and Ohnsorge, 2019; Checherita-Westphal and Domingues Semeano, 2020; Constitution of the Republic of Poland, 1997; Public Finance Act, 2009; Ministry of Finance of Poland, 2024, 2025; European Commission, 2026).

These endpoint assumptions warrant methodological caution. Debt sustainability analysis is an accounting exercise, deriving outcomes from assumptions on growth, interest, inflation, output gap closure, stock flow adjustments, and primary balances. The same accounting identity that fixes the 2036 endpoint also fixes the gap between the assumed and the debt-stabilising primary balance, so the endpoint and its implausibility are two readings of one set of inputs.

Within EU surveillance, these assumptions can effectively function as policy constraints, influencing prior guidance, assessment of fiscal plans, net expenditure ceilings, annual monitoring, and corrective pathways. Such practices risk circular reasoning: a pessimistic baseline could reinforce arguments for fiscal consolidation, while evaluating that consolidation might suppress public investment, thereby restricting growth feedback on the debt denominator. The scenario results presented below measure margins relative to this Commission surveillance construction. They do not imply that the baseline fiscal balances are domestically realistic, since by the baseline's own interest and growth assumptions those balances leave the debt ratio rising every year. Because the Commission baseline uses the Maastricht general government gross debt concept, later references to that baseline should not be read as applying Poland's national public debt rule unless the text states otherwise (Constitution of the Republic of Poland, 1997; Public Finance Act, 2009; European Commission, 2026). Figure 1 illustrates the Commission baseline debt trajectory before the paper adds any scenario-specific percentage point deviations from that baseline.

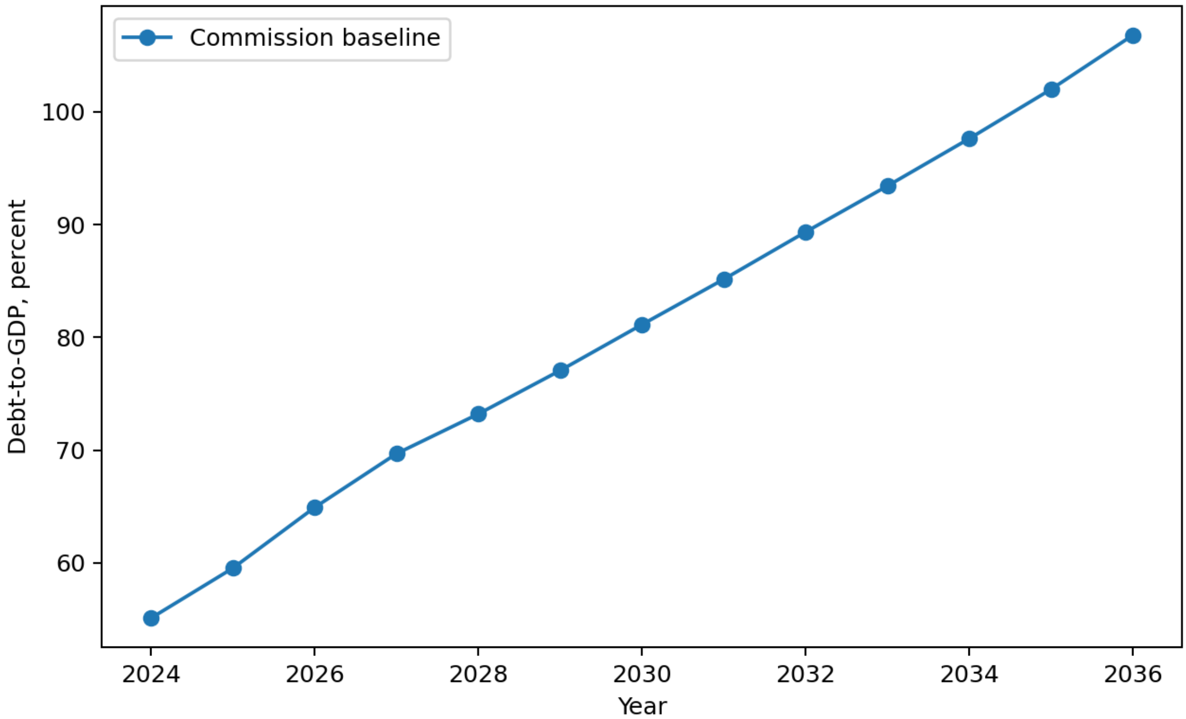

Figure 1: Baseline public debt path for Poland in the EU DSA application. The figure reports the debt-to-GDP ratio as a percent of GDP in the baseline series used by the paper before the public investment scenario is applied.

1.2 Critical assumptions in debt projections and consolidation evidence

Because these assumptions carry institutional weight, they need strong empirical support. Internal consistency within the debt projections does not by itself settle the policy concerns that follow. The empirical relationships invoked to justify consolidation efforts, debt thresholds, and expenditure reductions must themselves be reliable enough to bear the policy uses placed on them.

Dependence on critical assumptions extends beyond the Commission framework. Reinhart and Rogoff (2010) reported influential evidence linking elevated public debt with lower growth rates. Subsequent replication studies and threshold analyses, however, showed that these findings were highly sensitive to methodological decisions: data treatment methods, weighting schemes, sample selection, functional forms, and assumptions about the causal direction between debt and growth. This controversy shows the need for caution when interpreting debt ratios as rigid constraints on economic growth (Herndon, Ash and Pollin, 2013; Pescatori, Sandri and Simon, 2014; Panizza and Presbitero, 2013; Egert, 2013; Heimberger, 2021).

This debate over debt thresholds carries an initial caveat: causal inferences drawn from observed correlations between debt and growth remain highly sensitive to choices about data handling, functional specification, sample composition, and reverse causality assumptions. Read together with the evidence on expenditure consolidations discussed below, the constraint implies that empirical strategies which aggregate diverse fiscal instruments, lean on crisis-induced policy choices, or attribute macroeconomic outcomes to broad fiscal balance indicators cannot reliably demonstrate that reductions in public investment protect economic growth or reduce debt levels. A rigorous assessment instead has to identify an instrument-specific shock, estimate the resulting dynamic output response, and feed those dynamic effects explicitly into the debt denominator (Herndon, Ash and Pollin, 2013; Pescatori, Sandri and Simon, 2014; Panizza and Presbitero, 2013; Egert, 2013; Heimberger, 2021).

A similar evidentiary issue arises in the broader literature on expansionary austerity. Alesina, Favero and Giavazzi (2015) constructed multi-year fiscal adjustment plans, categorized each plan by its predominant tax or spending composition, and interpreted subsequent macroeconomic outcomes as evidence of the relative output costs of those classifications. The identification problem here is not specific to one study but runs through this entire line of work: aggregate plans are not fiscal instruments, and plan labels are not shocks. A plan classified as expenditure-based can combine public consumption, transfers, public investment, procurement deferral, tax measures, and implementation changes, while the decision to consolidate often occurs under debt stress, weak growth, market pressure, or external surveillance.

The originators of this literature concede the gap themselves. Alesina and Ardagna (2010) acknowledge that "there can still be an endogeneity issue related to the occurrence of fiscal adjustments and expansions, because, in principle, discretionary policy choices of fiscal authorities can be affected by countries' macroeconomic conditions", and that the correlation between a successful adjustment and a falling debt ratio "is not perfect" because the numerator can drop faster than the denominator. Perotti (2011) reopens the flagship cases and finds that in three of the four canonical episodes, Ireland, Finland and Sweden, growth was driven by exports and exchange-rate depreciation rather than by internal demand, with falling interest rates and wage moderation doing the work, a configuration that "is unlikely" to recur "in the present circumstances, with low inflation and interest rates close to zero". The causal reading of plan-level evidence is therefore exposed to the narrative reappraisal, inverse-propensity-weighting, and reverse-causality critiques of Guajardo, Leigh and Pescatori (2014), Jorda and Taylor (2016), and Breuer (2019). Plan-level identification cannot establish the dynamic causal effect of an expenditure instrument on output or debt, and least of all the effect of cutting public investment.

Borys, Ciżkowicz and Rzońca (2013) advance the expansionary-austerity claim across the demand components on which that claim depends. Their New Member State panel identifies fiscal impulses through four aggregate procedures and reports that expenditure-based fiscal adjustments are accompanied by accelerated private investment and export growth, stronger wage moderation, and improved cost competitiveness, whereas expenditure-based fiscal stimuli decelerate private investment and export growth and weaken competitiveness (Borys, Ciżkowicz and Rzońca, 2013). The authors read the small estimated effect on aggregate GDP not as a refutation but as confirmation, since in their account expenditure cuts shift growth toward investment, exports, and the cost channel.

That result cannot support claims about cutting public investment for three reasons, and the strongest version of each reason is one the authors themselves state. First, the fiscal impulses rest on aggregate balance measures and an action-based proxy that mix expenditure instruments together and never isolate public investment. The identifiers are the cyclically-adjusted primary balance (the budget balance net of interest, purged of the part of the deficit that moves automatically with the business cycle, abbreviated CAPB), the underlying balance (the CAPB further corrected for one-off transfers), an impulse derived from the simplified growth-accounting rule associated with von Hagen, and a reduced action-based indicator that records only the sign of legislated fiscal action rather than its size. The authors qualify the CAPB themselves, noting that it "should be used with caution, as noticed by the IMF (2010)" (Borys, Ciżkowicz and Rzońca, 2013). The wider action-based dataset literature reaches the same diagnosis for this whole family of indicators, treating cyclically-adjusted balance measures as contaminated by cyclical measurement error and by reverse causality, since rising output mechanically lowers an expenditure-to-GDP ratio rather than the reverse (Guajardo, Leigh and Pescatori, 2014; Breuer, 2019; Adler, Allen, Ganelli and Leigh, 2024). None of these identifiers can attribute a measured effect to public investment, because public investment is never separated from public consumption, transfers, or procurement within the aggregate.

Second, the timing of consolidation episodes is exposed to the selection problem that the authors concede they cannot remove. Their panel runs on a short sample, and they write that controlling for endogeneity through fiscal impulses "may be insufficient to fully eliminate the endogeneity bias", that the natural remedy, an instrumental-variables estimator, is barred because "the severe bias of this estimator when applied to short samples prevent us from using it", and that the results "should be treated with caution because of the estimation problems typical for panel data models, notably in the case of the limited number of available observations" (Borys, Ciżkowicz and Rzońca, 2013). The same selection mechanism documented for the wider literature applies here: weak growth, fiscal stress, market pressure, or external constraints can determine when a government consolidates and also shape the macroeconomic path that follows (Jorda and Taylor, 2016; Breuer, 2019).

Third, every channel the authors propose runs through wage moderation, cost competitiveness, private investment, and exports, none of which is public capital formation, and none of their specifications carries an estimated output path into a debt-to-GDP denominator. The result speaks to how expenditure composition correlates with private-sector demand in a transition-economy panel; it does not estimate the dynamic effect of a public-investment cut on output, and it computes no debt ratio. The authors close on the same note, conceding that "the conclusions drawn on their basis should be taken with caution" (Borys, Ciżkowicz and Rzońca, 2013). On the question this paper addresses, the policy implication is the opposite once public investment is isolated as an instrument and its persistent output response is integrated into debt dynamics: cutting public investment is estimated to raise, not lower, Poland's debt-to-GDP ratio relative to baseline.

The Borys et al. case is therefore not an isolated objection but an instance of a broader problem in consolidation evidence. Three limitations matter for the use of this literature in a public-investment debt exercise. The first is plan endogeneity. Consolidation plans are not genuine policy shocks; they typically arise when debt stress, unfavorable market circumstances, sluggish economic growth, or external constraints are already present. Treatment status then correlates directly with the macroeconomic conditions in place at the plan's implementation, so plan-level coefficients confound the underlying motivations for fiscal adjustment with observed economic outcomes. Guajardo, Leigh and Pescatori (2014) show this directly: replacing conventional cyclically adjusted primary balance measures with narratively identified consolidation episodes reverses the primary conclusions of the expansionary austerity literature, and expenditure-based consolidations turn out to be contractionary. Jorda and Taylor (2016) reach a comparable conclusion through inverse propensity score weighting of consolidation episodes; the ostensibly neutral output responses associated with consolidations primarily reflect selection bias rather than causal impacts. Breuer (2019) points to a reverse-causality problem inherent in conventional cyclically adjusted approaches, and shows that adjusting for this bias can remove or even reverse the perceived expansionary effects of fiscal adjustments.

The second limitation concerns aggregation. Plan-level treatments across the expansionary-austerity literature, including Alesina, Favero and Giavazzi (2015), classify multi-year consolidation plans by their predominant tax or expenditure components. This classification is not designed to isolate the effects of specific expenditure instruments, and the limitation holds for the whole composition-based research programme rather than for any single application. Consolidation episodes frequently encompass simultaneous actions such as restricting current spending, adjusting transfers, delaying procurement, reducing infrastructure outlays, or revising implementation schedules, and each of these instruments differs in timing, persistence, import leakage, and effect on the public capital stock. Breuer (2019) sharpens the same point at the measurement level: the cyclically adjusted primary balance used throughout this literature "fails to correct for cyclical effects in the expenditure-GDP-ratio", so that estimated expansionary effects are biased upward by reverse causality once output movements feed back into the ratio. Evidence on the composition of consolidation therefore does not separate one expenditure instrument from another when evaluating the macroeconomic consequences of fiscal adjustment. For the question studied here this gap is decisive, because public investment is precisely the instrument that aggregate composition labels submerge, which is why the analysis below measures movements in public investment directly and traces their dynamic output responses (Alesina, Favero and Giavazzi, 2015; Breuer, 2019; Ardanaz, Cavallo, Izquierdo and Puig, 2021).

The third limitation concerns persistence in the output response and its feedback into debt dynamics. Blanchard and Leigh (2013) show that European consolidation episodes produced systematic growth forecast errors in line with underestimated fiscal multipliers. DeLong and Summers (2012), Fatas and Summers (2018), and Gechert, Horn and Paetz (2019) explain why this matters for debt dynamics: when the output contraction persists, the weaker debt denominator and endogenous budgetary responses can offset any immediate improvement in the primary balance. Cugnasca and Rother (2015) reach a similar conclusion for the EU consolidation experience, with realised output responses that exceeded those implied by the low multiplier assumptions used in surveillance frameworks. Carbonari, Farcomeni, Maurici and Trovato (2024) add a related causal reappraisal of consolidation announcements: spending cut announcements can generate negative indirect output effects once mediation channels are accounted for. The point is therefore narrow but important: plan labels are not sufficient evidence that expenditure cuts, especially investment cuts, improve debt outcomes.

For Poland, these methodological implications enter a concrete surveillance setting through distinct EU fiscal surveillance mechanisms. The medium term fiscal structural plan establishes a net expenditure path monitored through annual progress reports, while deficit correction operates through an excessive deficit procedure embedded in a Council adjustment process. Fiscal surveillance therefore shapes both the pace and composition of Poland's fiscal adjustments. Macroeconomic evaluations for Poland must look beyond aggregate fiscal indicators to inflation, external balance, balance of payments conditions, domestic fiscal rules, and the investment component of public expenditure. Grodzicki, Mozdzen and Zygmuntowski (2022) support the fiscal-rule part of this argument: they criticise numerical fiscal aggregates as incomplete policy criteria and stress the macroeconomic context in which debt, growth, inflation, and external balance must be assessed. The investment-composition part comes from the public-investment and fiscal-rule literature reviewed below, while the external-transmission part is grounded in the multiplier evidence on openness, exchange-rate regimes, import content, and production structure.

1.3 Fiscal multipliers and instrument composition

This paper focuses on one instrument, public investment. Higher public investment raises current demand, supports public capital formation, and expands future productive capacity. Cutting it consolidates the primary balance mechanically, but it also lowers output and shrinks the public capital stock. The literature on budget composition and fiscal rule design gives further reason to treat public investment on its own. Investment projects are easier to delay than transfers, wages, or entitlement programmes, and their economic costs are often less visible at the moment of budgetary decision making (Breunig and Busemeyer, 2012; Borge and Hopland, 2015; Pereira and Pinho, 2006; OECD, 2011). That makes public investment a particularly exposed adjustment item: projects can be postponed, scaled back, or trimmed through annual allocations. The design of fiscal rules also bears on whether investment spending is protected or curtailed during adjustment episodes (Ardanaz et al., 2020). The central analytical question is therefore how debt projections change when public investment expansion and public investment consolidation include estimated growth feedbacks, rather than relying only on a single short run multiplier applied directly to the structural primary balance.

The Commission's Debt Sustainability Monitor 2025 uses a common fiscal multiplier of 0.6 in the relevant fiscal policy scenarios. A 1 percentage point improvement in the structural primary balance reduces actual GDP growth by 0.6 percentage points in the same year, while potential growth is assumed to stay unchanged (European Commission, 2026). The multiplier reaches the debt ratio through two channels. Lower GDP mechanically raises the debt-to-GDP ratio by shrinking the denominator, and weaker activity shifts the cyclical component of the budget balance, which then feeds into the primary balance path. The Debt Sustainability Monitor framework also applies a specified output gap closure rule. The Commission recognises that multipliers vary with structural characteristics, cyclical conditions, monetary policy settings, policy instruments, and the composition of adjustment, and it lets member states justify different values in their own plans. Even so, the operational simplification remains a common value used for prior guidance across countries.

An instrument specific question requires more than a common multiplier. Public investment, public consumption, transfers, and taxes have different dynamic effects. Multipliers vary by openness, exchange rate regime, public capital stock, fiscal position, monetary accommodation, and horizon (Ilzetzki, Mendoza and Vegh, 2013; Ramey and Zubairy, 2018; Cloyne, Jorda and Taylor, 2023). Evidence for Poland and for cross country samples already documents this heterogeneity. Table 1. Polish multiplier evidence used to motivate instrument specific estimation.

Source	Fiscal measure	Reported multiplier values
Ministry of Finance of Poland, Medium Term Fiscal Structural Plan 2025-2028	NEMPF effective consolidation multiplier and one year category multipliers	Effective consolidation multiplier: 0.449 in 2026, 0.498 in 2027, 0.528 in 2028; one year category multipliers: total expenditure 0.788, public investment 1.493.
Haug, Jedrzejowicz and Sznajderska (2019)	Government spending in Poland	Impact 0.70; four quarters 1.13; eight quarters 1.46; peak 1.61 after fourteen quarters.
Sznajderska (2025)	Fiscal policy shocks in Poland	Impact spending multiplier 1.25; long run cumulative spending multiplier 1.04; impact tax multiplier -1.18.
Haug, Lyziak and Sznajderska (2025)	Local projection IV government spending in Poland	Peak cumulative linear multiplier 1.53 after six quarters; pre-Covid peak 1.38 after eight quarters.

Note: The negative tax multiplier follows the standard sign convention for a positive tax shock: higher taxes reduce output.

Table 2. Cross country and European multiplier evidence used to motivate instrument specific estimation.

Source	Fiscal measure or conditioning dimension	Reported multiplier values
Ilzetzki, Mendoza and Vegh (2013)	Government consumption, high income countries	Impact 0.39; long run 0.66.
Ilzetzki, Mendoza and Vegh (2013)	Openness and exchange rate regime	Closed economies: impact 0.61, long run 1.10; open economies: impact -0.077, long run -0.46; fixed exchange rates: long run about 1.4; flexible exchange rates: long run about -0.69.
Ilzetzki, Mendoza and Vegh (2013)	Pure government investment shocks	High income countries: impact 0.39, long run 1.5; developing countries: impact 0.57, long run 1.6.
IMF Working Paper 2019/289	Public investment and initial public capital stock	Public investment: impact 0.15, two years 0.80; low initial public capital: two years 2.15; high initial public capital: two years 0.15.

Source	Fiscal measure or conditioning dimension	Reported multiplier values
Ciaffi, Deleidi and Capriati (2024)	Government spending in OECD countries; linear and high public debt states	Linear model: impact 0.72, five year 0.87; high debt cases: impact 0.64 to 0.80, five year 1.51 to 1.56.
Gechert and Will (2012)	Meta regression by fiscal instrument	Reported means: general public spending 1.0; public investment 1.2; transfers 0.4; taxes 0.6.
Saccone (2022)	Public investment in European countries	Total public investment: 0.979 on impact and 2.056 at horizon 6; economic affairs investment: 0.859 on impact and 2.336 at horizon 6.

The tables show that Polish official planning and the wider multiplier literature already distinguish fiscal instruments, reporting values that vary by horizon, instrument, and conditioning environment. This public investment evidence is what motivates estimating that instrument separately and carrying its output path into the debt assessment.

The paper is organised as follows. Section 2 presents the data, fiscal shock construction, and local projection methodology. Section 3 reports the estimated response trajectories. Section 4 applies them to public investment expansion and consolidation scenarios for Poland. Section 5 concludes. The appendices are organised sequentially to build from raw inputs to final results: Appendix A defines the state variables and reports the diagnostic screening; Appendix B reports the full cumulative response paths; Appendix C reports the annual debt-to-GDP projection paths; and Appendix D details the underlying local projection coefficient estimates, standard errors, and p-values.

2 Data and empirical strategy

2.1 Local projection framework

Local projections estimate dynamic responses by running a separate regression at each horizon after a policy intervention or shock. The outcome dated $t + h$ is regressed on the shock dated t conditional on controls, and the resulting sequence of horizon-specific coefficients traces the impulse response. The approach follows Jorda (2005) and the later local projection literature, which recovers impulse responses directly and does not impose the full dynamic restrictions of a vector autoregression (Jorda and Taylor, 2024). For fiscal multiplier analysis the method is well suited, since the estimate depends on the shock definition, the fiscal instrument, the horizon, and the conditioning information set (Ramey and Zubairy, 2018; Ramey, 2019).

The public investment shock is identified before the local projection regressions are estimated, because a change in public investment is not automatically a policy shock. Public investment growth may reflect current activity, common funding cycles, interest rate conditions, project completion, or planned procurement rather than an exogenous fiscal impulse. The recursive system therefore isolates the unexplained movement in public investment under a timing restriction, and the local projections then estimate how that identified innovation propagates across horizons. This separation between shock identification and response estimation follows the fiscal shock literature: Blanchard and Perotti (2002) identify fiscal shocks before tracing their macroeconomic effects; Jorda (2005) estimates horizon responses directly once the treatment variable is defined; Ramey (2019) and Ramey and Zubairy (2018) stress that multiplier estimates depend on the shock measure; and Ciaffi, Deleidi and Di Domenico (2024) first identify public investment shocks and then insert them into local projections. The same separation carries over here to an annual EU27 panel and to Polish state evaluation.

The first step estimates a system containing public investment growth, public consumption growth, output growth, and the short term interest rate. Public investment is ordered first in the recursive timing structure. This follows the fiscal shock literature, which treats public investment as relatively predetermined within the year because large investment decisions are shaped by multiannual political, administrative, and procurement processes (Ciaffi, Deleidi and Di Domenico, 2024; Ciaffi, Deleidi and Mazzucato, 2024). Under this timing restriction, output, public consumption, and the interest rate may respond within the same year to the public investment movement, while public investment does not respond contemporaneously to those variables. What remains of public investment growth, the part the system leaves unexplained, is then used as the public investment shock in the local projections.

The source data are annual EU27 series beginning in 2004. Where a variable is available through 2025, the 2025 observation enters the source and state-profile work. The local projection estimation uses horizon-specific samples, so the eighth-year sample uses 2004-2017 observations while shorter horizons use later years. A Poland only annual regression would provide too few observations since 2004 to estimate controlled dynamic responses through the eighth year, and too little variation to assess how fiscal transmission changes with country characteristics. The EU27 panel supplies variation across countries and time while keeping the estimation inside the institutional setting relevant for Poland's fiscal surveillance. Country fixed effects absorb persistent cross country differences, and year fixed effects absorb common yearly disturbances. The linear EU27 specification estimates a common public investment response conditional on fixed effects and lagged controls. Poland is evaluated later through interactions between the public investment shock and Poland's observed state values. For each horizon $h \in \{0, \dots, 8\}$, the baseline linear local projection is

$$y_{i,t+h} = \alpha_i^h + \tau_t^h + \beta_h s_{i,t}^{GI} + \gamma_h s_{i,t}^{GC} + \Gamma_h' W_{i,t-1} + u_{i,t+h}^h.$$

Here i indexes EU27 member states and t indexes years. The dependent variable $y_{i,t+h}$ is the scaled outcome at horizon h: real GDP in output regressions and public investment spending in spending regressions. The term $s_{i,t}^{GI}$ is the identified public investment shock, while $s_{i,t}^{GC}$ is the identified public consumption shock included as a fiscal control. The lagged controls match the fiscal, output, and monetary variables used in the recursive shock-identification system, following the practice of conditioning local projections on pre shock dynamics in the relevant information set (Jorda, 2005; Ciaffi, Deleidi and Di Domenico, 2024).

The vector $W_{i,t-1}$ contains lagged public investment growth, lagged public consumption growth, lagged output growth, and the lagged short term interest rate. The terms α_i^h and τ_t^h denote country and year fixed effects. The coefficient β_h is therefore the common linear EU27 response to the public investment shock at horizon h, conditional on fixed effects, the public consumption shock, and lagged controls. The linear specification imposes the same marginal response across the panel once these controls are in place. To see how that response changes with country characteristics, the paper also estimates state dependent local projections. The approach follows the interaction logic in Cloyne, Jorda and Taylor (2023), where the effect of a policy intervention can vary with observed information from before the shock year. It sits alongside the fiscal multiplier literature, in which responses vary with macroeconomic conditions and policy settings (Ramey and Zubairy, 2018). Here the state variables are continuous predetermined characteristics rather than binary recession expansion indicators.

For each individual transmission channel c, the state dependent specification is

$$y_{i,t+h} = \alpha_i^h + \tau_t^h + \beta_{c,h} s_{i,t}^{GI} + \theta_{c,h}(s_{i,t}^{GI} z_{c,i,t-1}) + \lambda_{c,h} z_{c,i,t-1} + \gamma_{c,h} s_{i,t}^{GC} + \Gamma'_{c,h} W_{i,t-1} + u_{i,t+h}^h.$$

The scalar $z_{c,i,t-1}$ is the lagged standardised value of one predetermined state variable. Because the state variable is standardised, the coefficient $\beta_{c,h}$ gives the public investment response for a country at the sample mean of that channel. The interaction coefficient $\theta_{c,h}$ shows how the response changes when a country's state value departs from that mean. The term $\lambda_{c,h}$ lets the same state variable enter the outcome equation directly, separately from its interaction with the public investment shock. The public consumption shock, lagged controls, and fixed effects are the same as in the linear specification. Poland's evaluation comes from applying its observed standardised value for channel c to the estimated interaction structure. Poland's evaluated response is therefore generated within the state dependent fixed effects local projection by the mean state coefficient $\beta_{c,h}$ together with the interaction term $\theta_{c,h} z_{c,PL,t-1}$.

In short, the identified shocks enter horizon-specific local projections as regressors. This shock-identification-plus-local projection structure follows Ciaffi, Deleidi and Di Domenico (2024), and the present paper adapts it to an EU27 annual panel and to the evaluation of Polish state variables.

The state variables used in the interaction terms are lagged and standardised on the EU27 panel, as defined in Section 2.2. Appendix A reports the state variable definitions and the eighth year output interaction selection rule used for the Polish evaluations.

Section 2.4 sets out the main limitations of this empirical design after the state-variable definitions and the channel-retention criterion.

2.2 State variables

This section defines the country characteristics that condition how fiscal shocks propagate within the EU27 panel. The four structural state variables are investment import content, public debt, household liquidity position, and real PPP income level. Each one is defined before estimation, tied to a specific economic mechanism, and measured on harmonised cross country data (Ilzetzki, Mendoza and Vegh, 2013; Huidrom, Kose, Lim and Ohnsorge, 2019; Kaplan, Violante and Weidner, 2014; Cloyne, Jorda and Taylor, 2023).

In the regression framework, a state variable is a predetermined country characteristic that interacts with fiscal shocks and summarises the setting in which fiscal transmission occurs. It is recorded before the shock year and is not part of the response trajectory after the shock. This distinction follows Cloyne, Jorda and Taylor's (2023) separation between treatment, conditioning state, and dynamic propagation. It also matches fiscal position research that treats initial fiscal conditions as possible determinants of multiplier strength rather than as outcomes of fiscal policy (Huidrom, Kose, Lim and Ohnsorge, 2019). The investment import content data come from the OECD TiVA database released in 2025. In the source used here, official measurements of domestic value added shares in gross fixed capital formation run through 2022, so the common TiVA state window is 2004 to 2022.

The operational definitions are standardised before Section 2.3 reports the baseline Polish profile. Table 2a reports each variable's measurement, observation count, standardisation moments, and Poland's value before and after standardisation. Investment import content uses official OECD TiVA observations through 2022. Public debt and real PPP income use Eurostat observations through 2025. Household liquidity position uses Eurostat through 2025 where available; Ireland's missing 2025 financial-account row affects only calculations that require that household financial-account input.

Table 2a. State-variable measurement and Polish profile, Panel A: definitions.

State variable	Measurement	Non-missing observations	Source window used
Investment import content	One minus OECD TiVA domestic value added share of gross fixed capital formation	513	Official TiVA through 2022; Poland profile uses the latest official TiVA value
Public debt	Maastricht gross debt, percent of GDP	594	Eurostat through 2025
Household liquidity position	Negative household net financial worth divided by GDP	593	Eurostat through 2025 where available; Ireland lacks the 2025 financial-account input
Real PPP income level	Log real GDP per capita in 2020 PPS terms	594	Eurostat through 2025

Table 2a continued. State-variable measurement and Polish profile, Panel B: standardisation and Poland values.

State variable	EU27 mean	EU27 standard deviation	Poland value	Poland standardised value
Investment import content	0.429	0.101	0.413	-0.161
Public debt	62.658	36.311	59.700	-0.081
Household liquidity position	-1.138	0.598	-0.793	0.576
Real PPP income level	10.237	0.380	10.265	0.074

Investment import content enters because an investment shock raises domestic output only to the extent that the spending is met by domestic value added rather than imports. The measure uses the OECD TiVA domestic value added share of gross fixed capital formation, and import content is one minus that share. A higher value therefore means that more of the investment demand leaks into imports instead of domestic value added. This state links the paper's investment channel to the openness mechanism in the fiscal multiplier literature (Ilzetzki, Mendoza and Vegh, 2013; Cacciatore and Traum, 2020).

Public debt enters as a fiscal-position state. Ilzetzki, Mendoza and Vegh (2013) identify public indebtedness as a country characteristic relevant for fiscal multiplier heterogeneity, while Huidrom, Kose, Lim and Ohnsorge (2019) link fiscal positions to interest rate and risk premium channels. Debt therefore has two roles. In Section 2, the public debt ratio is a pre-shock fiscal-position characteristic conditioning the estimated output response. In Section 4, the future debt-to-GDP path is the outcome generated by the scenario analysis.

Public debt is measured as Maastricht general government gross debt relative to GDP. The Maastricht debt ratio is used here only as a pre-shock fiscal-position state. This conditioning state can influence fiscal transmission through risk premia and the interest rate and growth dynamics that govern debt sustainability (Blanchard, 2019; Huidrom, Kose, Lim and Ohnsorge, 2019), and it enters the domestic and EU fiscal-rule assessments framing the adjustment path (Grodzicki, Mozdzen and Zygmuntowski, 2022; Ministry of Finance of Poland, 2024; European Commission, 2026). Section 1.1 distinguishes the EU Maastricht measure from Polish national public debt rules.

Household liquidity position is measured from financial accounts as the negative of household financial assets net of liabilities, divided by GDP. The numerator excludes non-financial assets, including housing, and the measure is transformed so that higher values indicate a weaker liquidity position relative to GDP. The variable captures the household financial-buffer channel, since fiscal shocks may propagate differently when households have limited net financial capacity to smooth consumption after income fluctuations. Kaplan, Violante and Weidner (2014) provide the microeconomic distinction between liquid resources and illiquid wealth that motivates treating liquid household resources as relevant for consumption smoothing. Bernardini, De Schryder and

Peersman (2017) show that household leverage conditions the transmission of fiscal shocks through balance sheet positions. Krajewski and Pilat (2025) give the Polish household-finance context in which liquidity constraints and financial buffers are empirically relevant. The paper's EU27 measure is therefore a comparable macro financial-accounts state, not a direct household survey measure of liquid deposits.

Real PPP income level is defined as the logarithm of real GDP per capita, expressed in 2020 purchasing power standard terms. Its role is to capture heterogeneity in development levels, including differences in income levels, market structures, relative-price environments, and broader macroeconomic conditions. Ilzetzki, Mendoza and Vegh (2013) explicitly examine multiplier differences between high income and developing economies, while Huidrom, Kose, Lim and Ohnsorge (2019) similarly link multiplier variation to structural factors and macro financial conditions.

These four state variables make up the paper's state-variable set because each maps directly to a theoretically interpretable channel for public-investment transmission. Investment import content captures import leakage from investment demand, public debt captures the fiscal-position channel, household liquidity position captures the household financial-buffer channel, and real PPP income level captures development-level heterogeneity. The choice is made before estimation and is deliberately parsimonious. Cloyne, Jorda and Taylor (2023) note the value of keeping the state space interpretable, while Ilzetzki, Mendoza and Vegh (2013) and Huidrom, Kose, Lim and Ohnsorge (2019) motivate multiplier heterogeneity through clearly defined country characteristics. The framework therefore evaluates four pre-specified individual transmission channels rather than searching across an unrestricted set of conditioning variables.

Unemployment and the output gap belong to the cyclical state-dependence literature rather than to the slower moving structural conditioning variables used here. Auerbach and Gorodnichenko (2012, 2013) and Ramey and Zubairy (2018) study fiscal multiplier differences between slack and expansion states, a distinct cyclical approach to state-dependent fiscal transmission. This study instead works with slower moving structural heterogeneity across EU27 economies. The narrower scope also reduces contemporaneous state-endogeneity concerns, since unemployment and the output gap can move at the same time as fiscal shocks and output responses (Cloyne, Jorda and Taylor, 2023). The analysis therefore focuses on structural conditioning variables, while recession-expansion asymmetries remain a separate analytical design.

2.3 Channel retention criterion and diagnostics

Section 2.2 defined the four channels and the reasons for evaluating them one at a time. This subsection sets out the criterion that decides which channels are carried into the Polish scenario. Within the finite annual EU27 panel, the design gives the coefficients a direct interpretation as distinct economic channels.

Each channel evaluation must meet several conditions: the local projection must be estimable, exhibit acceptable numerical stability, include Poland within the observed EU27 data range, have viable output and spending denominators, and show an eighth year output interaction at $p \leq 0.10$. The eighth year is selected as the diagnostic horizon because it directly informs the cumulative response comparison and the debt translation.

The design applies empirical discipline to state-dependent local projections. Relevant conditioning dimensions can matter for the completeness of interaction effects, while estimating several interaction dimensions at once can weaken empirical support, produce unstable estimates, and place too much weight on a limited number of country-year observations (Cloyne, Jorda and Taylor, 2023). The paper therefore separates the a priori definition of the state-variable set from the statistical decision to carry a channel into the debt scenario. The resulting estimates should be read as individual channel evaluations, not as a model of simultaneous interaction among all macroeconomic states.

Four individual channel evaluations are conducted. Investment import content, household liquidity position, and public debt each satisfy the eighth year output-interaction criterion, with p values of 0.002, 0.003, and 0.080. Real PPP income level, however, does not meet this criterion (p value of 0.461) and is therefore excluded from the Polish scenario projections. Table 2b details the retained channel evaluations that inform the primary response and subsequent debt analysis.

Table 2b. Retained Polish channel evaluations carried into the main scenario.

Retained Polish channel evaluation	State variable	Eighth year output interaction p value	Polish standardised state value
Investment import content	Investment import content	0.002	-0.161
Household liquidity position	Household liquidity position	0.003	0.576
Public debt	Public debt	0.080	-0.081

Notes: In the investment import content evaluation, Poland is slightly below the EU27 mean import content value for gross fixed capital formation. In the household liquidity position evaluation, Poland is above the EU27 mean of the transformed household liquidity state, indicating weaker net financial worth relative to GDP under the paper's sign convention. In the public debt evaluation, Poland is slightly below the EU27 mean Maastricht public debt ratio in the mixed-window state profile.

Appendix A reports the state variable definitions and the statistical screening results for all four individual channels. Real PPP income level belongs to the pre-specified set of state variables, but it is not retained because its eighth year output interaction does not satisfy the stated criterion. Section 3 therefore reports empirical response paths for the linear EU27 benchmark and for the three retained Polish channel evaluations. Appendix D reports the coefficient-level estimation output behind these paths: horizon-specific shock coefficients, interaction coefficients, standard errors, p values, observation counts, country counts, year ranges, fixed effects, and the covariance estimator. These diagnostics describe coefficient-level precision and are distinct from any full-path confidence interval for the debt translation.

2.4 Limitations of the empirical design

The empirical strategy has several limitations that should be read together with the response paths in Section 3 and the debt translation in Section 4. First, the local projections estimate conditional dynamic responses to recursively identified public-investment shocks. The identifying variation thus relies on the maintained within-year timing restriction that orders public investment before contemporaneous output, public consumption, and the short term interest rate. Local projections are transparent and flexible, yet they can be less smooth and less efficient than a correctly specified dynamic system, and horizon-specific estimates should not be read as a fully specified structural model of fiscal policy (Jorda, 2005; Jorda and Taylor, 2024).

Second, state dependence is evaluated through pre-specified individual channels. This maintains interpretability of the state space, but relying on a single dimension of state dependence may obscure complexities arising from interactions among multiple state variables. The evaluations for investment import content, public debt, household liquidity position, and real PPP income should thus be understood as separate conditional paths rather than as part of a joint model where all macroeconomic states interact simultaneously. This approach follows the caution of Cloyne, Jorda and Taylor (2023) that interpretable state spaces are valuable but do not eliminate the possibility of richer interaction structures.

Third, interactions among state variables are not jointly estimated. This restriction is substantive, as import leakage, fiscal position, household liquidity, and development level may interact during actual transmission. Jointly estimating all interactions in a finite EU27 annual panel would weaken empirical discipline, increase instability, and disproportionately weight limited country-year observations. Their omission means reported paths should be viewed as partial channel evaluations rather than as a complete interaction surface for fiscal transmission (Cloyne, Jorda and Taylor, 2023; Goncalves, Herrera, Kilian and Pesavento, 2023).

Fourth, the sample defines the inference scope. The panel begins in 2004, using annual EU27 variation, since a Poland-only annual regression would lack sufficient observations to estimate controlled dynamic responses through the eighth year. Polish paths are therefore EU27 panel estimates evaluated at Polish state values rather than Poland-specific time-series estimates. Support and diagnostic checks mitigate extrapolation risk but do not transform panel heterogeneity into a country-specific natural experiment (Cloyne, Jorda and Taylor, 2023).

Fifth, the horizon structure is finite. The eighth-year diagnostic horizon is used because it feeds the cumulative response comparison and debt translation, yet it remains a deliberate design choice. The reported local projections do not estimate effects beyond the eighth year and do not imply zero effects after this horizon. Horizon-specific leads also shorten the usable sample at longer horizons, so the eighth-year evidence relies on the shorter 2004-2017 estimation window, while shorter horizons draw on more recent observations. This limitation affects the interpretation of persistence but does not alter the reported response values, the stated retention rule, or the scenario arithmetic.

3 Results

3.1 EU27 panel benchmark and Polish evaluations within the EU27 panel

The results section first reports cumulative output responses to public investment, because the debt exercise in Section 4 uses the estimated output paths as inputs.

For each horizon h, $K_Y(h)$ denotes the cumulative real GDP response through horizon h to the identified public investment shock. The corresponding $K_G(h)$ denotes the cumulative movement in public investment spending generated by the same shock. The distinction matters because Section 4 states the scenario as a policy action equal to one percentage point of GDP in each of three years, while the empirical model estimates the output and spending paths that follow the identified public investment innovation. $K_G(h)$ translates the estimated shock into the scale of the stated fiscal action; $K_Y(h)$ carries the resulting output feedback into the debt calculation. The two series need not move together. Public investment spending can rise on impact and then partly unwind, while output can persist through demand, capacity and balance sheet channels.

The ratio $K_Y(h)/K_G(h)$ is therefore the cumulative output response per cumulative unit of public investment spending, the multiplier concept closest to the fiscal multiplier literature (Ilzetzki, Mendoza and Vegh, 2013; Ramey and Zubairy, 2018; Ciaffi, Deleidi and Di Domenico, 2024). This ratio is the relevant comparison object for a multiplier such as the Commission's 0.6, but the comparison only orients: the Commission value is a one year GDP growth response to a structural primary balance change, whereas $K_Y(h)/K_G(h)$ is an eighth year cumulative investment spending ratio. Table 3 reports $K_Y(h)$, $K_G(h)$, and $K_Y(h)/K_G(h)$ through the eighth annual horizon for the EU27 panel benchmark, the three Polish debt-scenario channel evaluations carried into the main scenario, and their equal weight arithmetic average.

Table 3. Eighth year cumulative output and spending responses for public investment.

Path	Evaluation basis	$K_Y(8)$	$K_G(8)$	$K_Y(8)/K_G(8)$	Role
EU27 benchmark	Fixed effects panel, no state interactions	2.23	0.78	2.87	EU27 reference
Investment import content	Poland's TiVA investment import content profile	1.80	0.73	2.47	Import leakage evaluation
Household liquidity position	Poland's household financial-account liquidity profile	2.24	0.77	2.91	Household liquidity position evaluation
Public debt	Poland's Maastricht public debt profile	1.78	0.75	2.37	Fiscal position evaluation

Path	Evaluation basis	$K_Y(8)$	$K_G(8)$	$K_Y(8)/K_G(8)$	Role
Equal weight average	Arithmetic average of the three Polish debt-scenario paths	1.94	0.75	2.59	Section 4 summary path

Notes: The output and spending responses are cumulative response units under the public investment shock normalization. Positive output values mean output above the no shock path. The ratio divides the eighth year cumulative output response by the eighth year cumulative spending response. Rounded ratios correspond to Appendix B, Table B.3.

The EU27 panel benchmark gives an eighth year cumulative output response of 2.23. This is the panel average response after conditioning on country and year fixed effects and the lagged controls in the linear specification. Because the linear specification imposes a common slope on the public investment shock, the estimate serves as an EU27 benchmark, not as a Poland specific response.

The investment import content evaluation for Poland yields an eighth-year cumulative output response of 1.80. Poland's evaluated import content sits slightly below the EU27 panel mean, with a raw value of 0.413 against the EU27 mean of 0.429 and a standardised value of -0.161. The comparison with the EU27 benchmark is not a mechanical, single-variable exercise. Within the investment import content specification, Poland's below-average state value raises the eighth-year output response relative to the same specification evaluated at the EU27 mean, while the EU27 benchmark reflects a distinct linear panel response without state interactions. The Polish investment import content trajectory runs above the EU27 benchmark initially, then falls below it from the third horizon onward. The result is a 2036 debt endpoint that is less self defeating than the EU27 benchmark, though still self defeating under a cut. This pattern fits the trade multiplier literature, which treats import leakage as composition dependent rather than governed by a simple openness rule: fiscal transmission depends on the domestic value added content of demand, the relative import content of public versus private expenditure, production structure, financing, and invoicing conditions (Ilzetzki, Mendoza and Vegh, 2013; Cacciatore and Traum, 2020; Crespo Cuaresma and Glocker, 2023).

The household liquidity position Polish evaluation gives an eighth year cumulative output response of 2.24. Here Poland is evaluated through the household sector's net financial balance sheet position rather than through a direct credit flow measure. The state captures a financial accounts balance sheet channel: fiscal shocks can transmit differently when household financial assets net of liabilities are weaker relative to GDP (Kaplan, Violante and Weidner, 2014; Bernardini, De Schryder and Peersman, 2017).

The public debt Polish evaluation gives an eighth year cumulative output response of 1.78. Here the conditioning state is the pre-shock Maastricht public debt ratio. Poland's evaluated public debt state is slightly below the EU27 mean, with a standardised value of -0.081. This path captures heterogeneity by fiscal position in the estimated transmission of public investment shocks, and it is distinct from the later debt-to-GDP outcome in Section 4.

The equal weight average across the three Polish debt-scenario evaluations is 1.94 in the eighth year.

3.2 Persistence and shape of the output responses across horizons

Section 3.1 reported eighth year cumulative output responses for the EU27 panel benchmark, the three Polish debt-scenario evaluations, and their equal weight average. Those values are endpoints of full cumulative response paths. This subsection follows $K_Y(h)$, the cumulative output response at horizon h, from impact through the eighth year.

Figure 2 plots the full sequence. At each horizon, $K_Y(h)$ is the cumulative real GDP response under the paper's public investment shock normalization. The figure includes the EU27 benchmark, the three Polish debt-scenario evaluations, and the equal weight Polish average.

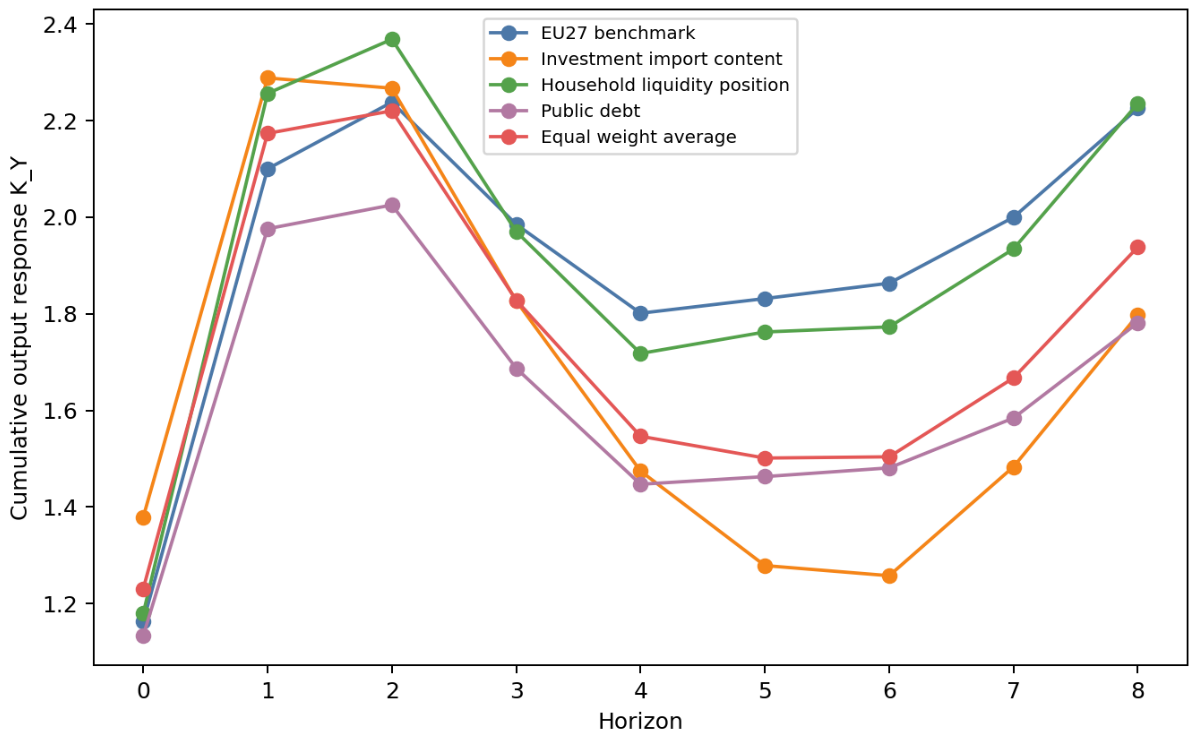

Figure 2: Cumulative output response paths $K_Y(h)$ from impact through the eighth year. The figure plots the EU27 benchmark, the investment import content Polish evaluation, the household liquidity position Polish evaluation, the public debt Polish evaluation, and the equal weight Polish average.

The estimated paths stay positive throughout the reported horizon. They rise strongly in the first two years, moderate across the middle horizons, and rise again by the eighth year. The sign matters for the scenario application: for an equal public investment cut, the same response paths imply a persistent output loss rather than a transitory impact effect. In the eighth year, the investment import content Polish evaluation reaches 1.80. The household liquidity position evaluation reaches 2.24, and the public debt evaluation reaches 1.78. Their equal weight average is 1.94.

Table 4 reports selected horizons from the same paths, and Appendix B reports the full values at each horizon.

Table 4. Selected values of $K_Y(h)$.

Empirical path	$h = 0$	$h = 2$	$h = 5$	$h = 8$
EU27 panel benchmark	1.16	2.24	1.83	2.23
Polish evaluation based on investment import content	1.38	2.27	1.28	1.80
Polish evaluation based on household liquidity position	1.18	2.37	1.76	2.24
Polish evaluation based on public debt	1.13	2.03	1.46	1.78
Equal weight average across the three Polish debt-scenario paths	1.23	2.22	1.50	1.94

The selected horizons show that the EU27 benchmark and the equal weight Polish average remain close in the eighth year, while the three Polish debt-scenario evaluations differ in magnitude. In the eighth year, the investment import content response is 1.80, the household liquidity position response is 2.24, and the public debt response is 1.78. The average summarizes the Polish debt-scenario evaluations, but it does not remove the difference among the import leakage, household liquidity, and fiscal-position channels.

The main implication is persistence over the observed local projection horizon. The estimated response extends beyond the impact year: cumulative output responses stay positive through the eighth year, and the later values are still economically material. For an expansion, persistence raises output during the years in which the debt ratio is assessed. For an equal cut, reversing the sign of the public investment action turns the same response path into a sustained output shortfall. The hysteresis literature gives the economic reason this persistence is not cosmetic: demand shortfalls can reduce investment, weaken labour market attachment, and lower future productive capacity (DeLong and Summers, 2012; Fatas and Summers, 2018; Engler and Tervala, 2016; Gechert, Horn and Paetz, 2019).

4 Application to Polish public investment scenarios

4.1 Scenario design

Polish public investment sits where national development planning, EU fiscal surveillance, and discretionary political decisions about project implementation meet. Poland's medium term fiscal structural plan is built around a defined net expenditure path extending to 2028. Official Polish documents place Poland under the excessive deficit procedure since July 2024. Evaluation under this procedure turns on adherence to the recommended expenditure path, and it tracks the progress of reforms and investments explicitly tied to EU priorities (Ministry of Finance of Poland, 2024, 2025; European Commission, 2026). The institutional challenge has two sources: the overall size of the fiscal balance and the composition of budgetary adjustments. It sharpens when multiannual investment commitments extend across several budget cycles under a surveillance framework that closely examines expenditure growth and debt trajectory forecasts.

Centralny Port Komunikacyjny shows how a programme can keep formal continuity while its annual investment intensity falls sharply. The revised programme retains CPK in official planning and extends the active framework to 2024-2032, yet the real public-investment trajectory is smaller than under the 2023 programme. The headline envelope decreases from PLN 155.1 billion to PLN 131.7 billion, the State Treasury engagement limit declines from PLN 66.2 billion to PLN 62.9 billion, the timeline is prolonged by two years, and maximum annual State Treasury bond engagement decreases from over PLN 13 billion to approximately PLN 11.5 billion. These changes are more than presentational. They cut the scale and the annual public-financing intensity of the programme while preserving its formal title and official continuity. Within the surveillance context set out above, this is the mechanism by which public investment can be curtailed without explicitly cancelling a programme: the project remains formally intact, but its annual budgetary and implementation intensity is reduced through a smaller envelope, lower Treasury involvement, lower annual bond engagement, and postponement (Council of Ministers of Poland, 2023, 2024b; Centralny Port Komunikacyjny, 2025).

The Polish Nuclear Power Programme shows the same pattern: formal continuity with material deferral. The declared goal of constructing two nuclear power plants with a combined capacity of approximately 6 to 9 GWe remains valid, yet the investment timeline has slipped. The 2020 programme scheduled the first unit to become operational in 2033. The 2025 update sets commercial operation of the first reactor for 2036, with subsequent units targeted for 2037 and 2038. Resolution No. 66 of 24 June 2024 revised Appendix 3 on implementation expenditure, returning the second plant to a staged preparatory phase that includes location assessment, technology and contractor selection, business model design, financing arrangements, and ownership structure.

Konin makes the shift concrete. The previously identified Pątnów route, with the foreign partner KHNP, has given way to an open preparatory phase led by PGE, in which Konin is now only one of several coal-region locations under analysis alongside Bełchatów, Kozienice and Połaniec. The 2022 official documents detailed cooperation among PGE, ZE PAK and KHNP for Pątnów; later reports indicated KHNP's withdrawal from the Polish nuclear project (Ministry of State Assets of Poland, 2022; Yonhap News Agency, 2025; Pulse, 2025; Ministry of Energy of Poland, 2025). Moving from a committed implementation route with a designated partner back to renewed site analysis is itself a substantial investment cut behind formal programme continuity, alongside delayed investment decisions and a slower annual expenditure trajectory. More broadly, the nuclear programme shows that declared strategic continuity does not rule out significant restraint in the public-investment path when milestones, implementation routes, and annual expenditure profiles are deferred (Polish Nuclear Power Programme, 2020; Council of Ministers of Poland, 2024a; Ministry of Industry of Poland, 2025; Ministry of State Assets of Poland, 2022; Ministry of Energy of Poland, 2025).

Fiscal policy is evaluated here as a path, not as a sequence of independent annual shocks. Three strands of the literature support this. First, cumulative and present value multipliers compare output responses with spending responses over a horizon, rather than only an impact coefficient (Ilzetzki, Mendoza and Vegh, 2013; Ramey and Zubairy, 2018; Ramey, 2019; Huidrom, Kose, Lim and Ohnsorge, 2019; Saccone, 2022). Second, fiscal plans are normally implemented over time, so their effects on budgets and GDP should be evaluated as paths rather than as isolated one year events (Alesina, Favero and Giavazzi, 2015; Ramey, 2019). Third, debt outcomes depend on how spending, output, the primary balance, and the debt-to-GDP denominator interact (Fatas and Summers, 2018; Ciaffi, Deleidi and Di Domenico, 2024). The policy translation used below is this paper's scenario implementation: three annual public investment actions of one percentage point of GDP in 2028, 2029, and 2030, combined with the estimated spending and output response paths.

Let $\sigma = 1$ denote the expansion case, and let $\sigma = -1$ denote the cut case. The programmed annual action vector is

$$a_s = \begin{cases} \sigma, & s = 0, 1, 2, \\ 0, & s \geq 3, \end{cases}$$

where $s = 0, 1, 2$ correspond to 2028, 2029, and 2030. A single 2028 action generates the spending path $P_G^{(1)}(h) = \sigma K_G(h)$ and the output path $P_Y^{(1)}(h) = \sigma K_Y(h)$. The three year programme then adds the delayed responses from the three programmed actions:

$$P_G^{(3)}(h) = \sum_{s=0}^{\min(h,2)} a_s K_G(h-s), \qquad P_Y^{(3)}(h) = \sum_{s=0}^{\min(h,2)} a_s K_Y(h-s).$$

The 2031 spending effect is therefore $a_0 K_G(3) + a_1 K_G(2) + a_2 K_G(1)$, with $a_3 = 0$. The nonzero value after 2030 is not a fourth annual fiscal action. It is the continuing dynamic response to the three actions already introduced. The one percentage point figure is the annual policy scale within a multiannual investment framework, not the full dynamic spending trajectory.

Both cases use the same years, scale, and debt accounting horizon. The cut case reverses the sign of the annual investment action. It changes the discretionary primary balance impulse and the estimated output response, while keeping the accounting environment constant.

The first debt calculation applies the medium term debt equation from the Debt Sustainability

Monitor framework. Here the debt-to-GDP ratio is carried forward by the interest and growth term, the primary balance, and stock flow adjustments (European Commission, 2026). The public investment action changes the discretionary primary balance path, and the estimated output response changes the GDP feedback entering the debt ratio. The calculation keeps the baseline projection path as the accounting environment, but replaces the relevant growth feedback with the estimated public investment response.

The second debt calculation is the direct debt-to-GDP local projection path. It estimates the debt-to-GDP response to the same public investment shock and applies that estimated debt ratio response to the same three annual actions. A cut initially increases the primary balance, yet the associated output loss and subsequent debt ratio response may still raise the debt ratio at the endpoint. An expansion initially lowers the primary balance, yet the associated output gain and resulting debt ratio response may lower the debt ratio at the endpoint. This calculation is consistent with debt and multiplier studies that evaluate fiscal policy through its effects on the debt-to-GDP ratio, and it parallels research linking government investment shocks to debt outcomes (DeLong and Summers, 2012; Fatas and Summers, 2018; Ciaffi, Deleidi and Di Domenico, 2024).

4.2 2036 debt-to-GDP impact across specifications

This subsection reports the 2036 debt-to-GDP margins for the public investment scenario defined in Section 4.1. Each table entry is a percentage point difference from the baseline debt-to-GDP ratio in 2036. Positive values mean the debt-to-GDP ratio is higher than baseline, negative values mean it is lower.

The table reports point scenario translations rather than joint confidence intervals. A joint confidence interval would have to combine uncertainty from the local projection response paths, the K_G scaling from shock units to fiscal actions, the direct debt response, and the Commission baseline accounting inputs. This application keeps these components visible separately. The absence of a joint confidence interval is therefore a limitation of the debt translation, not a claim of statistical precision for the final debt margins.

Table 5 reports the endpoint margins.

Table 5. 2036 debt-to-GDP margins relative to baseline.

Empirical path	Expansion, institutional debt equation	Expansion, direct debt-to-GDP local projection path	Cut, institutional debt equation	Cut, direct debt-to-GDP local projection path
EU27 panel benchmark	-7.1	-3.5	7.7	3.5
Polish evaluation based on investment import content	-4.1	-1.7	4.3	1.7
Polish evaluation based on household liquidity position	-6.8	-3.5	7.3	3.5
Polish evaluation based on public debt	-3.4	-2.7	3.7	2.7
Equal weight average across the three retained Polish channel evaluations	-4.8	-2.6	5.1	2.6

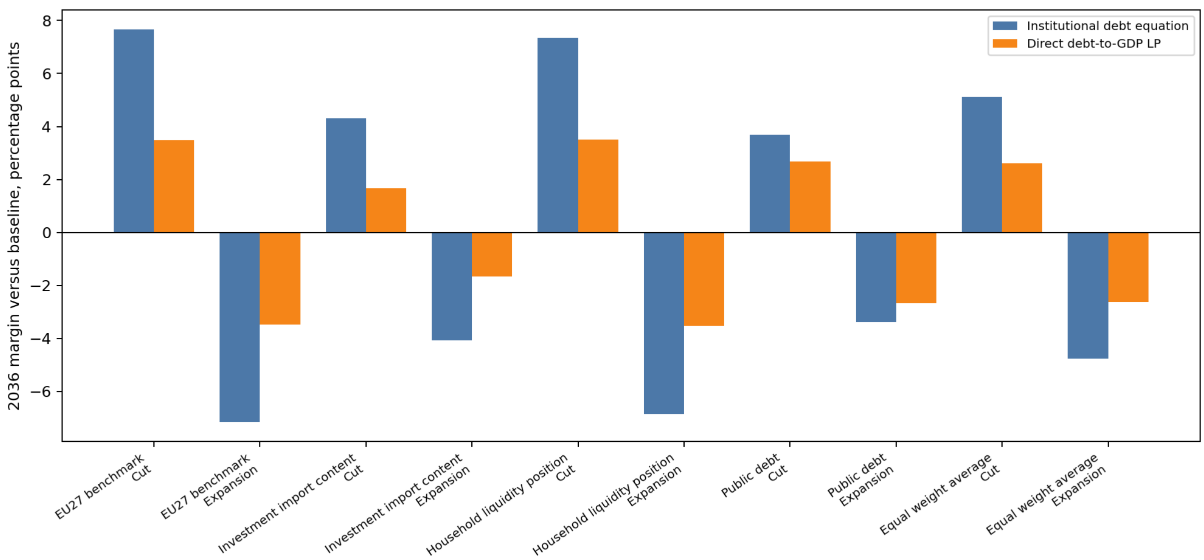

Figure 3: 2036 debt-to-GDP margins under expansion and cut scenarios. The figure reports both the institutional debt equation and the direct debt-to-GDP local projection path; positive values mean debt above baseline and negative values mean debt below baseline.

Figure 3 places the two scenario directions and the two debt calculations on the same footing. Because the baseline debt-to-GDP ratio rises over 2028-2036, the question that matters is whether each public investment scenario leaves the debt ratio above or below this baseline by 2036. Public investment expansions end below the baseline in 2036, while symmetric public investment cuts end above it. The shared endpoint pattern follows from output persistence, which enters the debt calculation directly. Persistent output gains after investment expansions lower the debt ratio, while sustained output losses after symmetric cuts raise it through the debt-to-GDP calculation.

The EU27 panel benchmark establishes a comparative standard for the analysis. A three year public investment expansion reduces the 2036 debt-to-GDP ratio by 7.1 percentage points according to the institutional debt equation and by 3.5 percentage points according to the direct debt-to-GDP local projection path. These outcomes are measured at the 2036 endpoint, eight years after the first action and six years after the final action. The corresponding symmetric public investment cut increases the debt ratio by 7.7 percentage points under the institutional debt equation and by 3.5 percentage points under the direct projection path.

In the Polish evaluation based on investment import content, the debt response is smaller than in the EU27 benchmark, but it has the same direction. The public investment expansion reduces the 2036 debt-to-GDP ratio by 4.1 percentage points under the institutional debt equation and by 1.7 percentage points under the direct debt-to-GDP local projection path. The public investment cut raises the debt ratio by 4.3 percentage points and 1.7 percentage points, respectively.

The Polish evaluation based on household liquidity position gives larger 2036 margins than the investment import-content evaluation. Under the institutional debt equation, a public investment expansion reduces the 2036 debt-to-GDP ratio by 6.8 percentage points, while the corresponding cut increases it by 7.3 percentage points. Under the direct debt-to-GDP path, the same expansion reduces the ratio by 3.5 percentage points, and the corresponding cut increases it by 3.5 percentage points.

The public debt evaluation gives the smallest institutional endpoint among the retained Polish channel evaluations, but it keeps the same sign. The expansion reduces the 2036 debt-to-GDP ratio by 3.4 percentage points under the institutional debt equation and by 2.7 percentage points under the direct debt-to-GDP local projection path. The corresponding cut increases the debt ratio by 3.7 percentage points and 2.7 percentage points.

The equal weight average across the three Polish evaluations reinforces the individual specification-level results. Under the institutional debt equation, the public investment expansion reduces the 2036 debt-to-GDP ratio by 4.8 percentage points; under the direct debt-to-GDP local projection path, it reduces the ratio by 2.6 percentage points. The corresponding public investment cut increases the debt ratio by 5.1 percentage points and 2.6 percentage points, respectively. The averaged analysis therefore confirms that public investment cuts do not produce debt improvements by 2036.

Taken together, the reported specifications show that public investment cuts consistently leave the 2036 debt-to-GDP ratio above baseline. Among the retained Polish channel evaluations, cut margins range from 3.7 to 7.3 percentage points under the institutional debt equation and from 1.7

to 3.5 percentage points under the direct debt-to-GDP local projection path. The expansion side of the same scenario tells the matching story: across the EU27 benchmark, the retained Polish channel evaluations and their equal weight average, public investment expansions consistently place the 2036 debt-to-GDP ratio below baseline under both debt calculation methods. The pattern is consistent with the self defeating consolidation literature: when output effects persist, the GDP channel can overturn the apparent fiscal gain from public investment cuts (DeLong and Summers, 2012; Fatas and Summers, 2018).

5 Conclusion

This paper examined public investment as both an instrument of development policy and an object of fiscal surveillance. Annual EU27 local projections estimate responses to public investment shocks and evaluate Poland from structural characteristics observed before the shock year. The main scenario uses three retained individual-channel evaluations specific to Poland: investment import content, household liquidity position and public debt. Under the paper's public investment shock normalization, cumulative eighth year output responses remain positive: the EU27 benchmark is 2.23, the Polish evaluations are 1.80 for investment import content, 2.24 for household liquidity position and 1.78 for public debt, and the equal weight Polish average is 1.94.

The debt application translates these outcomes into a medium term public investment scenario for Poland. The same three annual actions, each equal to 1 percentage point of GDP in 2028, 2029 and 2030, are applied with opposite signs for expansion and consolidation. In the 2036 endpoint results, the expansion reduces the debt-to-GDP ratio relative to baseline under both debt calculation methods, while symmetric investment cuts increase it. Across the three retained Polish channel evaluations, endpoint margins for cuts range from 3.7 to 7.3 percentage points under the institutional debt equation and from 1.7 to 3.5 percentage points under the direct debt-to-GDP local projection path. This is the self defeating public investment consolidation mechanism the paper identifies: the primary balance gain from investment cuts can be offset by the debt-ratio effect of persistent output losses. Hysteresis matters through the persistence of estimated responses over the medium term horizon. When investment cuts hold output down, they can weaken the GDP denominator and related budget feedbacks, and this overrides apparent accounting improvements.

These findings matter for EU fiscal surveillance because debt projections are sensitive to multiplier assumptions, output persistence and the composition of fiscal adjustments. A common short run multiplier can make investment cuts look more effective as a consolidation measure than they are once instrument-specific output paths enter the calculation. When projected debt paths, expenditure ceilings and progress assessments shape national fiscal choices, empirically based output responses for public investment change the estimated debt outcomes of both expansion and consolidation measures substantially.

For Polish public investment strategy, the analysis speaks directly to programmes implemented over multiple budget cycles. Transport and nuclear investment programmes can remain in official documents even as their timing, annual funding envelope, staging or implementation route changes.

Fiscal assessments must therefore look beyond programme adoption and read the annual budget decisions, including whether real investment content is expanded, preserved, narrowed or deferred. Treating public investment mainly as an expenditure item available for consolidation understates its role in development policy and misstates its debt implications.

In Poland's context, the persistence of estimated output responses feeds directly into debt outcomes. Public investment should therefore not be treated as an easy budgetary margin for meeting the Commission Debt Sustainability Monitor baseline used in this paper: cutting investment risks worsening the debt ratio, while increasing investment may improve it when medium term output effects are strong enough.

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Appendix A. Data, State Variables, and Channel Disclosure

This appendix defines the state variables used in the Polish evaluations and reports the statistical screening results for the four individual transmission channels considered in the paper. These state variables are predetermined country characteristics, measured before the public investment shock and standardised across the EU27 annual panel. The data window uses Eurostat observations through 2025 where available, keeps Ireland wherever the required inputs exist, and limits the missing Irish financial-account observation to calculations that require household financial-account data. Investment import content is measured with the latest official OECD TiVA observation, which is 2022 in the public data used here.

Let $x_{i,t}$ denote a raw state value for country i in year t. The standardised state is

$$z^x_{i,t} = \frac{x_{i,t} - \bar{x}}{s_x},$$

where $\bar{x}$ and s_x are the EU27 panel mean and standard deviation over the stated source window.

Table A.1a. State variable formulas and source classes.

State variable	Formula before standardisation	Source class
Investment import content	$m^I_{i,t} = 1 - DVA^{GFCF}_{i,t}/100$	OECD TiVA domestic value added shares for gross fixed capital formation
Public debt	$d_{i,t} = 100 \times D^M_{i,t}/Y_{i,t}$	Eurostat general government Maastricht debt
Household liquidity position	$w_{i,t} = -(FA_{i,t} - FL_{i,t})/Y_{i,t}$	Eurostat financial accounts and nominal GDP
Real PPP income level	$q_{i,t} = \log(GDPPC^{PPS2020}_{i,t})$	Eurostat national accounts

Table A.1b. State variable standardisation and Polish profile.

State variable	EU27 mean	EU27 standard deviation	Poland value	Poland standardised value
Investment import content	0.429	0.101	0.413	-0.161
Public debt	62.658	36.311	59.700	-0.081
Household liquidity position	-1.138	0.598	-0.793	0.576
Real PPP income level	10.237	0.380	10.265	0.074

Table A.2a. Source institutions and exact data codes for state variables.

Input	Primary source	Source variable and data codes
Investment import content	OECD TiVA, 2025 release	DSD_TIVA_MAINSH@DF_MAINSH(1.1), indicator GFCF_VA_SH, total activity _T, counterpart D, unit PT_GFCF_VA
Household liquidity position	Eurostat financial accounts and national accounts	Financial accounts from nasa_10_f_bs, sector S14_S15, na_item=F, finpos=ASS, LIAB, co_nco=NCO, unit MIO_EUR; GDP denominator from nama_10_gdp, B1GQ, CP_MEUR
Real PPP income	Eurostat national accounts	nama_10_pc, B1GQ, CP_PPS_EU27_2020_HAB and CLV_I20_HAB
Public debt state	Eurostat government deficit, debt and associated data	gov_10dd_edpt1, sector S13, na_item=GD, unit PC_GDP

Table A.2b. State-variable windows, coverage, and units.

Input	Years used	Coverage	Unit
Investment import content	2004 to 2022	EU27; Poland profile uses official 2022 TiVA	Import-content share of gross fixed capital formation
Household liquidity position	2004 to 2025 where available	EU27 except Ireland in the 2025 financial-account observation	Ratio to nominal GDP
Real PPP income	2004 to 2025	EU27	Log real PPP GDP per inhabitant
Public debt state	2004 to 2025	EU27	Percent of GDP

Table A.2c. State-variable transformations used in the manuscript.

Input	Transformation used in manuscript
Investment import content	$m^I_{i,t} = 1 - DVA^{GFCF}_{i,t}/100$; standardised on the EU27 TiVA panel
Household liquidity position	$w_{i,t} = -(FA_{i,t} - FL_{i,t})/Y_{i,t}$; standardised on the available EU27 panel
Real PPP income	$q_{i,t} = \log(PPS2020_{i,t} \times CLV20_{i,t}/100)$; standardised on the EU27 panel
Public debt state	Maastricht gross debt ratio; standardised on the EU27 panel

Table A.2d. Commission debt inputs and scenario source windows.

Input	Primary source	Years used	Coverage
Baseline debt ratio	European Commission Debt Sustainability Monitor 2025	2024 to 2036	Poland
Structural balance	European Commission Debt Sustainability Monitor 2025	2024 to 2036	Poland
Primary balance	European Commission Debt Sustainability Monitor 2025	2024 to 2036	Poland
Nominal growth assumptions	European Commission Debt Sustainability Monitor 2025 and scenario debt inputs	2025 to 2036	Poland
Interest and growth terms	European Commission Debt Sustainability Monitor 2025	2024 to 2036	Poland
Stock flow adjustment	European Commission Debt Sustainability Monitor 2025	2024 to 2036	Poland
Ageing costs	European Commission Debt Sustainability Monitor 2025	2024 to 2036	Poland
Public investment scenario action	Author scenario design	2028 to 2036	Poland

The following diagnostics explain how individual channel evaluations are screened before any Polish response path is reported. The screen first checks numerical estimability, then empirical support around Poland's state profile, denominator strength, and finally the eighth year output interaction used in the retention criterion.

For an individual channel c, let X_c be the local projection regressor matrix at the diagnostic horizon and let $z_{PL,c}$ be Poland's standardised state value. Numerical estimability is evaluated by

$$\operatorname{rank}(X_c) = k_c, \qquad \kappa_c = \sqrt{\frac{\lambda_{\max}(X_c' X_c)}{\lambda_{\min}(X_c' X_c)}}.$$

Empirical support is evaluated by the distance

$$D^2_{PL,c} = \frac{(z_{PL,c} - \bar{z}_c)^2}{\hat{\sigma}_c^2},$$

with the support p value computed from the corresponding reference distribution. The output interaction at $h = 8$ supplies the response-relevance criterion:

$$H_0 : \theta_{8,c} = 0.$$

The Polish scenario carries the individual channel evaluations that satisfy the numerical diagnostics, support diagnostics, denominator diagnostics, and the eighth year output-interaction criterion of $p \leq 0.10$. In this mixed-window run, investment import content, household liquidity position, and public debt satisfy the criterion and are retained. Real PPP income level does not satisfy the criterion and is not retained in the scenario projections.

Table A.3. Individual channel screening by eighth year output interaction.

Individual transmission channel	Output interaction p value	Criterion result	Scenario role
Investment import content	0.002	Criterion satisfied	Retained Polish channel evaluation
Household liquidity position	0.003	Criterion satisfied	Retained Polish channel evaluation
Public debt	0.080	Criterion satisfied	Retained Polish channel evaluation
Real PPP income level	0.461	Criterion not satisfied	Not retained in the Polish scenario

Notes: The statistical screening criterion is applied only to the four individual transmission channels. The equal weight Polish path in the main results averages the three retained channel evaluations: investment import content, household liquidity position, and public debt.

Notes: The condition number and maximum state correlation indicate the numerical behavior of the included regressors. The support p value and maximum absolute Polish z score describe the position of Poland's evaluated state profile within the EU27 distribution. The criterion table reports the diagnostic outcomes used before a path enters the response and debt application.

Appendix B. Additional Dynamic Responses

Appendix B gives the full numerical response paths behind the selected horizons reported in Section 3. Table B.1 reports the cumulative output response $K_Y(h)$ for horizons $h = 0, \dots, 8$ for the EU27 panel benchmark, the three Polish evaluations, and their equal weight average. Table B.2 reports the corresponding cumulative spending response $K_G(h)$ for the same paths. Table B.3 reports $K_Y(h)/K_G(h)$, the cumulative output response per cumulative unit of public investment spending. The ratio is calculated row by row from the unrounded output and spending paths before display rounding. It is a reference measure for comparison with the multiplier literature; the debt application uses $K_Y(h)$ and $K_G(h)$ as separate inputs.

Table B.1. Cumulative output response paths, $K_Y(h)$, horizons $h = 0, \dots, 8$.

Horizon	EU27 benchmark	Investment import content	Household liquidity position	Public debt	Equal weight average
0	1.16	1.38	1.18	1.13	1.23
1	2.10	2.29	2.26	1.98	2.17
2	2.24	2.27	2.37	2.03	2.22
3	1.99	1.83	1.97	1.69	1.83
4	1.80	1.47	1.72	1.45	1.55
5	1.83	1.28	1.76	1.46	1.50
6	1.86	1.26	1.77	1.48	1.50
7	2.00	1.48	1.93	1.58	1.67
8	2.23	1.80	2.24	1.78	1.94

Table B.2. Cumulative spending response paths, $K_G(h)$, horizons $h = 0, \dots, 8$.

Horizon	EU27 benchmark	Investment import content	Household liquidity position	Public debt	Equal weight average
0	1.00	1.00	1.00	1.00	1.00
1	1.01	1.01	1.03	1.01	1.02
2	0.84	0.84	0.85	0.83	0.84
3	0.72	0.70	0.72	0.69	0.70
4	0.69	0.64	0.69	0.65	0.66
5	0.69	0.63	0.70	0.66	0.66
6	0.67	0.60	0.68	0.65	0.64
7	0.71	0.65	0.72	0.69	0.69
8	0.78	0.73	0.77	0.75	0.75

Table B.3. Cumulative output to spending ratios, $K_Y(h)/K_G(h)$, horizons $h = 0, \ldots, 8$.

Horizon	EU27 benchmark	Investment import content	Household liquidity position	Public debt	Equal weight average
0	1.16	1.38	1.18	1.13	1.23
1	2.09	2.26	2.19	1.97	2.14
2	2.66	2.71	2.80	2.45	2.65
3	2.76	2.61	2.75	2.43	2.60
4	2.62	2.31	2.48	2.21	2.33
5	2.66	2.02	2.53	2.21	2.26
6	2.78	2.10	2.61	2.26	2.33
7	2.82	2.27	2.70	2.30	2.43
8	2.87	2.47	2.91	2.37	2.59

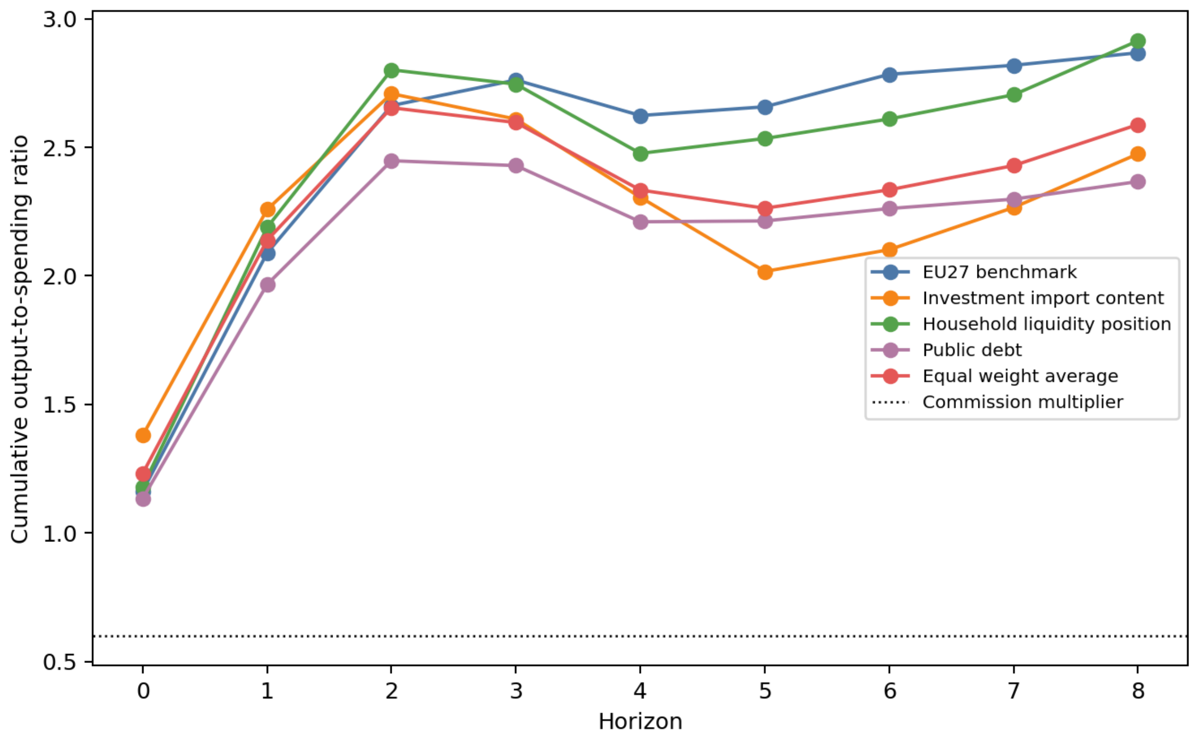

Figure 4: Cumulative output-to-spending ratio paths $K_Y(h)/K_G(h)$ from impact through the eighth year. The dotted horizontal line marks the Commission's 0.6 fiscal multiplier used in the Debt Sustainability Monitor scenario framework. The comparison is temporal at impact and cumulative thereafter; it is not an exact instrument match because the Commission value applies to a structural primary balance change, whereas the paths here are public-investment spending responses.

Notes: Values are cumulative responses to a public investment shock at horizon h under the paper's shock normalization. Table 3 reports the $h = 8$ entries from Tables B.1 to B.3. The EU27 benchmark comes from the linear panel specification with country and year fixed effects and supplies the common output and spending path used in the EU27 comparison. The Polish evaluations apply Poland's standardised state profile within the three Polish debt-scenario specifications carried into the main scenario. The equal weight average assigns one third to each Polish debt-scenario evaluation. Table B.3 reports $K_Y(h)$ divided by $K_G(h)$ for each path. Thus, at $h = 8$, the EU27 row reports $K_G(8) = 0.78$ and $K_Y(8)/K_G(8) = 2.87$; the ratio is computed from unrounded output and spending paths before display rounding.

Appendix C. Debt accounting decomposition

This appendix sets out the debt accounting framework used in the scenario application. It reports the 2036 endpoint margins together with the annual debt accounting decomposition. The application evaluates three annual public investment actions, each equal to 1 percentage point of GDP in 2028, 2029, and 2030. The expansion and cut cases use the same years and the same one percentage point scale. The expansion case treats the action as higher public investment, whereas the cut case applies the same action with the opposite sign. The institutional debt equation carries the debt ratio forward by applying the interest and growth factor to the preceding debt ratio, subtracting the primary balance, and adding stock flow adjustments. In this scenario application, the public investment action changes the discretionary primary balance path, while the estimated output response changes the GDP feedback component of the debt ratio calculation. In simplified notation, the recursion is

$$b_t = \frac{1 + i_t}{1 + g_t} b_{t-1} - pb_t + sfa_t,$$

where b_t is the debt-to-GDP ratio, i_t is the effective nominal interest rate on the debt stock, g_t is nominal GDP growth, pb_t is the primary balance as a share of GDP, and sfa_t is the stock flow adjustment. The scenario changes pb_t through the direct investment action and the cyclical primary balance feedback, while g_t changes through the estimated output response. The annual comparison is therefore

$$b_t^{scenario} = \frac{1 + i_t}{1 + g_t^{scenario}} b_{t-1}^{scenario} - pb_t^{scenario} + sfa_t, \qquad \Delta b_t = b_t^{scenario} - b_t^{baseline},$$

where the baseline path is the Commission debt path for the same year and Δb_t is the reported scenario margin. The annual tables compare the resulting scenario debt ratio with the Commission baseline debt ratio for the same year. They also display the output effect, direct primary balance effect, cyclical primary balance feedback, and primary balance path used in that comparison. The direct debt ratio path applies the estimated debt ratio response directly to the same three annual actions, outside the comprehensive institutional debt equation. Table C.1 presents the 2036 margins relative to the baseline debt ratio under both debt calculation approaches. The remaining tables provide the annual institutional debt equation decomposition for the EU27 panel benchmark, the three Polish debt-scenario evaluations, and their equal weight average. These scenario margins are differences relative to the baseline series used in the application and scenario translations of the estimated response paths. The Commission does not endorse them as official projections for alternative scenarios.

Table C.1. 2036 debt-to-GDP margins relative to baseline, percentage points.

Empirical path	Action	Institutional debt equation	Direct debt-to-GDP local projection path
EU27 benchmark	Expansion	-7.1	-3.5
EU27 benchmark	Cut	7.7	3.5
Investment import content	Expansion	-4.1	-1.7
Investment import content	Cut	4.3	1.7
Household liquidity position	Expansion	-6.8	-3.5
Household liquidity position	Cut	7.3	3.5
Public debt	Expansion	-3.4	-2.7
Public debt	Cut	3.7	2.7
Equal weight average	Expansion	-4.8	-2.6
Equal weight average	Cut	5.1	2.6

Notes: Negative values mean that the 2036 debt-to-GDP ratio is below baseline; positive values mean that it is above baseline. The institutional debt equation is the medium term debt accounting used in the application. The direct debt-to-GDP local projection path reports the estimated debt-ratio response to the same signed investment action.

Table C.b1. Annual debt accounting: EU27 benchmark, expansion - debt path and margins.

Year	Baseline debt ratio	Scenario debt ratio	Institutional debt margin	Direct debt ratio margin
2028	73.19	72.84	-0.35	-1.53
2029	77.05	75.65	-1.39	-4.05
2030	81.12	78.33	-2.80	-6.42
2031	85.14	81.27	-3.88	-6.94
2032	89.34	85.04	-4.30	-5.73
2033	93.42	88.83	-4.59	-3.40
2034	97.62	92.52	-5.09	-1.57
2035	101.99	96.03	-5.96	-1.58
2036	106.79	99.65	-7.14	-3.48

Table C.b2. Annual debt accounting: EU27 benchmark, expansion - output and primary balance.

Year	Output effect	Scenario nominal GDP growth	Direct primary balance effect	Cyclical primary balance feedback	Baseline primary balance	Scenario primary balance
2028	-1.16	6.65	-1.00	0.56	-3.88	-4.32
2029	-3.26	7.42	-2.01	1.57	-3.95	-4.39
2030	-5.50	7.45	-2.85	2.64	-3.99	-4.20
2031	-6.32	6.14	-2.57	3.04	-3.91	-3.44
2032	-6.02	5.12	-2.25	2.89	-4.02	-3.38
2033	-5.62	5.09	-2.09	2.70	-3.84	-3.24
2034	-5.50	5.42	-2.05	2.64	-3.86	-3.27
2035	-5.70	5.72	-2.07	2.73	-3.88	-3.21
2036	-6.09	5.61	-2.15	2.92	-3.89	-3.13

Table C.c1. Annual debt accounting: EU27 benchmark, cut - debt path and margins.

Year	Baseline debt ratio	Scenario debt ratio	Institutional debt margin	Direct debt ratio margin
2028	73.19	73.57	0.37	1.53
2029	77.05	78.58	1.53	4.05
2030	81.12	84.31	3.19	6.42
2031	85.14	89.55	4.40	6.94
2032	89.34	94.11	4.77	5.73
2033	93.42	98.40	4.98	3.40
2034	97.62	103.08	5.46	1.57
2035	101.99	108.37	6.38	1.58
2036	106.79	114.45	7.66	3.48

Table C.c2. Annual debt accounting: EU27 benchmark, cut - output and primary balance.

Year	Output effect	Scenario nominal GDP growth	Direct primary balance effect	Cyclical primary balance feedback	Baseline primary balance	Scenario primary balance
2028	1.16	4.20	1.00	-0.56	-3.88	-3.44
2029	3.26	3.00	2.01	-1.57	-3.95	-3.51
2030	5.50	2.74	2.85	-2.64	-3.99	-3.78
2031	6.32	4.40	2.57	-3.04	-3.91	-4.38
2032	6.02	5.76	2.25	-2.89	-4.02	-4.67
2033	5.62	5.96	2.09	-2.70	-3.84	-4.44
2034	5.50	5.68	2.05	-2.64	-3.86	-4.45
2035	5.70	5.30	2.07	-2.73	-3.88	-4.54
2036	6.09	4.78	2.15	-2.92	-3.89	-4.66

Table C.d1. Annual debt accounting: Investment import content, expansion - debt path and margins.

Year	Baseline debt ratio	Scenario debt ratio	Institutional debt margin	Direct debt ratio margin
2028	73.19	72.59	-0.60	-1.59
2029	77.05	75.10	-1.95	-4.12
2030	81.12	77.56	-3.56	-6.42
2031	85.14	80.74	-4.41	-6.72
2032	89.34	85.05	-4.29	-5.01
2033	93.42	89.70	-3.72	-1.82
2034	97.62	94.33	-3.29	0.75
2035	101.99	98.64	-3.36	0.84
2036	106.79	102.71	-4.08	-1.66

Table C.d2. Annual debt accounting: Investment import content, expansion - output and primary balance.

Year	Output effect	Scenario nominal GDP growth	Direct primary balance effect	Cyclical primary balance feedback	Baseline primary balance	Scenario primary balance
2028	-1.38	6.88	-1.00	0.66	-3.88	-4.22
2029	-3.67	7.61	-2.01	1.76	-3.95	-4.20
2030	-5.93	7.47	-2.85	2.85	-3.99	-3.99
2031	-6.38	5.77	-2.55	3.06	-3.91	-3.39
2032	-5.57	4.61	-2.18	2.67	-4.02	-3.53
2033	-4.58	4.51	-1.97	2.20	-3.84	-3.62
2034	-4.01	4.97	-1.87	1.93	-3.86	-3.81
2035	-4.02	5.53	-1.89	1.93	-3.88	-3.83
2036	-4.54	5.74	-1.98	2.18	-3.89	-3.69

Table C.e1. Annual debt accounting: Investment import content, cut - debt path and margins.

Year	Baseline debt ratio	Scenario debt ratio	Institutional debt margin	Direct debt ratio margin
2028	73.19	73.83	0.63	1.59
2029	77.05	79.18	2.13	4.12
2030	81.12	85.16	4.04	6.42
2031	85.14	90.11	4.97	6.72
2032	89.34	94.03	4.69	5.01
2033	93.42	97.37	3.95	1.82
2034	97.62	101.06	3.44	-0.75
2035	101.99	105.51	3.51	-0.84
2036	106.79	111.12	4.32	1.66

Table C.e2. Annual debt accounting: Investment import content, cut - output and primary balance.

Year	Output effect	Scenario nominal GDP growth	Direct primary balance effect	Cyclical primary balance feedback	Baseline primary balance	Scenario primary balance
2028	1.38	3.97	1.00	-0.66	-3.88	-3.54
2029	3.67	2.79	2.01	-1.76	-3.95	-3.69
2030	5.93	2.70	2.85	-2.85	-3.99	-3.99
2031	6.38	4.82	2.55	-3.06	-3.91	-4.42
2032	5.57	6.34	2.18	-2.67	-4.02	-4.52
2033	4.58	6.60	1.97	-2.20	-3.84	-4.07
2034	4.01	6.17	1.87	-1.93	-3.86	-3.91
2035	4.02	5.51	1.89	-1.93	-3.88	-3.92
2036	4.54	4.65	1.98	-2.18	-3.89	-4.09

Table C.f1. Annual debt accounting: Household liquidity position, expansion - debt path and margins.

Year	Baseline debt ratio	Scenario debt ratio	Institutional debt margin	Direct debt ratio margin
2028	73.19	72.82	-0.37	-1.51
2029	77.05	75.47	-1.58	-4.02
2030	81.12	77.94	-3.18	-6.31
2031	85.14	80.81	-4.34	-6.80
2032	89.34	84.75	-4.58	-5.56
2033	93.42	88.80	-4.62	-3.36
2034	97.62	92.70	-4.92	-1.60
2035	101.99	96.32	-5.67	-1.59
2036	106.79	99.95	-6.85	-3.51

Table C.f2. Annual debt accounting: Household liquidity position, expansion - output and primary balance.

Year	Output effect	Scenario nominal GDP growth	Direct primary balance effect	Cyclical primary balance feedback	Baseline primary balance	Scenario primary balance
2028	-1.18	6.67	-1.00	0.57	-3.88	-4.31
2029	-3.44	7.58	-2.03	1.65	-3.95	-4.33
2030	-5.80	7.58	-2.88	2.79	-3.99	-4.08
2031	-6.60	6.11	-2.59	3.17	-3.91	-3.34
2032	-6.06	4.89	-2.26	2.91	-4.02	-3.37
2033	-5.45	4.90	-2.11	2.62	-3.84	-3.33
2034	-5.25	5.35	-2.07	2.52	-3.86	-3.41
2035	-5.47	5.74	-2.09	2.63	-3.88	-3.34
2036	-5.94	5.69	-2.16	2.85	-3.89	-3.20

Table C.g1. Annual debt accounting: Household liquidity position, cut - debt path and margins.

Year	Baseline debt ratio	Scenario debt ratio	Institutional debt margin	Direct debt ratio margin
2028	73.19	73.59	0.39	1.51
2029	77.05	78.78	1.73	4.02
2030	81.12	84.75	3.62	6.31
2031	85.14	90.06	4.92	6.80
2032	89.34	94.40	5.06	5.56
2033	93.42	98.40	4.98	3.36
2034	97.62	102.86	5.24	1.60
2035	101.99	108.04	6.04	1.59
2036	106.79	114.13	7.33	3.51

Table C.g2. Annual debt accounting: Household liquidity position, cut - output and primary balance.

Year	Output effect	Scenario nominal GDP growth	Direct primary balance effect	Cyclical primary balance feedback	Baseline primary balance	Scenario primary balance
2028	1.18	4.18	1.00	-0.57	-3.88	-3.45
2029	3.44	2.83	2.03	-1.65	-3.95	-3.57
2030	5.80	2.59	2.88	-2.79	-3.99	-3.90
2031	6.60	4.44	2.59	-3.17	-3.91	-4.48
2032	6.06	6.03	2.26	-2.91	-4.02	-4.68
2033	5.45	6.18	2.11	-2.62	-3.84	-4.35
2034	5.25	5.76	2.07	-2.52	-3.86	-4.31
2035	5.47	5.28	2.09	-2.63	-3.88	-4.41
2036	5.94	4.69	2.16	-2.85	-3.89	-4.58

Table C.h1. Annual debt accounting: Public debt, expansion - debt path and margins.

Year	Baseline debt ratio	Scenario debt ratio	Institutional debt margin	Direct debt ratio margin
2028	73.19	72.87	-0.32	-1.28
2029	77.05	75.84	-1.20	-3.28
2030	81.12	78.82	-2.30	-4.88
2031	85.14	82.22	-2.92	-4.65
2032	89.34	86.53	-2.81	-3.12
2033	93.42	90.87	-2.55	-1.56
2034	97.62	95.10	-2.51	-0.96
2035	101.99	99.17	-2.82	-1.39
2036	106.79	103.41	-3.38	-2.68

Table C.h2. Annual debt accounting: Public debt, expansion - output and primary balance.

Year	Output effect	Scenario nominal GDP growth	Direct primary balance effect	Cyclical primary balance feedback	Baseline primary balance	Scenario primary balance
2028	-1.13	6.62	-1.00	0.54	-3.88	-4.34
2029	-3.11	7.29	-2.01	1.49	-3.95	-4.46
2030	-5.14	7.24	-2.83	2.47	-3.99	-4.36
2031	-5.69	5.87	-2.53	2.73	-3.91	-3.70
2032	-5.16	4.89	-2.18	2.48	-4.02	-3.72
2033	-4.60	4.93	-2.01	2.21	-3.84	-3.65
2034	-4.39	5.34	-1.97	2.11	-3.86	-3.72
2035	-4.53	5.66	-2.00	2.17	-3.88	-3.71
2036	-4.85	5.54	-2.10	2.33	-3.89	-3.66

Table C.i1. Annual debt accounting: Public debt, cut - debt path and margins.

Year	Baseline debt ratio	Scenario debt ratio	Institutional debt margin	Direct debt ratio margin
2028	73.19	73.53	0.34	1.28
2029	77.05	78.37	1.32	3.28
2030	81.12	83.76	2.63	4.88
2031	85.14	88.48	3.33	4.65
2032	89.34	92.48	3.14	3.12
2033	93.42	96.22	2.80	1.56
2034	97.62	100.36	2.74	0.96
2035	101.99	105.06	3.07	1.39
2036	106.79	110.47	3.68	2.68

Table C.i2. Annual debt accounting: Public debt, cut - output and primary balance.

Year	Output effect	Scenario nominal GDP growth	Direct primary balance effect	Cyclical primary balance feedback	Baseline primary balance	Scenario primary balance
2028	1.13	4.23	1.00	-0.54	-3.88	-3.42
2029	3.11	3.13	2.01	-1.49	-3.95	-3.44
2030	5.14	2.98	2.83	-2.47	-3.99	-3.62
2031	5.69	4.71	2.53	-2.73	-3.91	-4.11
2032	5.16	6.01	2.18	-2.48	-4.02	-4.32
2033	4.60	6.13	2.01	-2.21	-3.84	-4.04
2034	4.39	5.77	1.97	-2.11	-3.86	-4.00
2035	4.53	5.37	2.00	-2.17	-3.88	-4.04
2036	4.85	4.86	2.10	-2.33	-3.89	-4.12

Table C.j1. Annual debt accounting: Equal weight average, expansion - debt path and margins.

Year	Baseline debt ratio	Scenario debt ratio	Institutional debt margin	Direct debt ratio margin
2028	73.19	72.76	-0.43	-1.46
2029	77.05	75.47	-1.58	-3.81
2030	81.12	78.11	-3.01	-5.87
2031	85.14	81.26	-3.89	-6.06
2032	89.34	85.44	-3.89	-4.56
2033	93.42	89.79	-3.63	-2.25
2034	97.62	94.04	-3.58	-0.60
2035	101.99	98.04	-3.95	-0.71
2036	106.79	102.02	-4.77	-2.62

Table C.j2. Annual debt accounting: Equal weight average, expansion - output and primary balance.

Year	Output effect	Scenario nominal GDP growth	Direct primary balance effect	Cyclical primary balance feedback	Baseline primary balance	Scenario primary balance
2028	-1.23	6.72	-1.00	0.59	-3.88	-4.29
2029	-3.40	7.49	-2.02	1.63	-3.95	-4.33
2030	-5.63	7.43	-2.85	2.70	-3.99	-4.14
2031	-6.22	5.92	-2.56	2.99	-3.91	-3.48
2032	-5.59	4.80	-2.20	2.69	-4.02	-3.54
2033	-4.88	4.78	-2.03	2.34	-3.84	-3.53
2034	-4.55	5.22	-1.97	2.18	-3.86	-3.64
2035	-4.67	5.64	-1.99	2.24	-3.88	-3.63
2036	-5.11	5.65	-2.08	2.45	-3.89	-3.52

Table C.k1. Annual debt accounting: Equal weight average, cut - debt path and margins.

Year	Baseline debt ratio	Scenario debt ratio	Institutional debt margin	Direct debt ratio margin
2028	73.19	73.65	0.45	1.46
2029	77.05	78.78	1.73	3.81
2030	81.12	84.56	3.43	5.87
2031	85.14	89.55	4.41	6.06
2032	89.34	93.64	4.30	4.56
2033	93.42	97.33	3.91	2.25
2034	97.62	101.43	3.81	0.60
2035	101.99	106.20	4.21	0.71
2036	106.79	111.91	5.11	2.62

Table C.k2. Annual debt accounting: Equal weight average, cut - output and primary balance.

Year	Output effect	Scenario nominal GDP growth	Direct primary balance effect	Cyclical primary balance feedback	Baseline primary balance	Scenario primary balance
2028	1.23	4.12	1.00	-0.59	-3.88	-3.47
2029	3.40	2.92	2.02	-1.63	-3.95	-3.57
2030	5.63	2.76	2.85	-2.70	-3.99	-3.84
2031	6.22	4.65	2.56	-2.99	-3.91	-4.34
2032	5.59	6.13	2.20	-2.69	-4.02	-4.51
2033	4.88	6.30	2.03	-2.34	-3.84	-4.15
2034	4.55	5.90	1.97	-2.18	-3.86	-4.07
2035	4.67	5.39	1.99	-2.24	-3.88	-4.13
2036	5.11	4.73	2.08	-2.45	-3.89	-4.27

Notes: Output effects are reported relative to the baseline output level; negative values in the expansion case denote higher output relative to baseline in the debt accounting convention, while positive values in the cut case denote the corresponding output shortfall. Direct primary balance effects are the discretionary fiscal actions generated by the three year programme. Cyclical primary balance feedback records the endogenous budget response to the output movement. Institutional debt margins are the margins from the medium term debt equation; direct debt-to-GDP margins are the corresponding local projection debt-ratio margins estimated for the same signed investment action.

Appendix D. Estimation results

Appendix D reports the coefficient-level estimates for the 2025 mixed-window specification. It covers the linear EU27 benchmark, the four individual transmission channels evaluated at Poland's state profile, the horizon-by-horizon sample, and the bridge from incremental regression responses to cumulative output and public-investment paths. The tables expose the estimation surface used for the retained channel evaluations and the contextual real PPP income evaluation.

At each horizon, the linear EU27 benchmark estimates the response of the scaled outcome to an identified public-investment shock, controlling for the public-consumption shock, lagged macroeconomic variables, and country and year fixed effects. For outcome $m \in \{Y, G\}$ at horizon h, the benchmark equation is

$$x^m_{i,t+h} = \alpha^h_i + \tau^h_t + \beta^m_h s^{GI}_{i,t} + \gamma^m_h s^{GC}_{i,t} + \delta^{m\prime}_h W_{i,t-1} + u^m_{i,t+h},$$

where $x^Y_{i,t+h}$ denotes the scaled real GDP response, $x^G_{i,t+h}$ denotes the scaled public investment spending response, $s^{GI}_{i,t}$ is the identified public investment shock, $s^{GC}_{i,t}$ is the identified public consumption shock, and $W_{i,t-1}$ contains lagged controls. The parameters α^h_i and τ^h_t capture country and year fixed effects, respectively. The coefficient on the public-investment shock is therefore the horizon-specific output or spending response of interest; the remaining coefficients capture fixed effects and controls.

The channel evaluations allow the public-investment response to vary with one predetermined state characteristic at a time. For individual channel c, the standardised state variable $z_{c,i,t-1}$ enters both as a control and as an interaction with the public-investment shock:

$$x^m_{i,t+h} = \alpha^h_i + \tau^h_t + \beta^m_{c,h} s^{GI}_{i,t} + \theta^m_{c,h}(s^{GI}_{i,t} z_{c,i,t-1}) + \lambda^m_{c,h} z_{c,i,t-1} + \gamma^m_{c,h} s^{GC}_{i,t} + \delta^{m\prime}_{c,h} W_{i,t-1} + u^m_{i,t+h}.$$

The evaluated Polish incremental response at horizon h is computed on the same horizon-specific estimation sample used for the numerator. Let $\beta^{G0}_{c,h}$ and $\theta^{G0}_{c,h}$ denote the coefficients from the contemporaneous public-investment-spending scale equation, estimated on the horizon-h sample. The evaluated response is

$$r^m_c(h; PL) = \frac{\beta^m_{c,h} + \theta^m_{c,h} z_{c,PL}}{\beta^{G0}_{c,h} + \theta^{G0}_{c,h} z_{c,PL}}.$$

The linear EU27 benchmark is obtained from the benchmark equation without a state interaction. The denominator normalises the incremental response so that the immediate public-investment spending response equals one. The cumulative objects are

$$K_Y(h) = \sum_{q=0}^{h} r^Y(q), \qquad K_G(h) = \sum_{q=0}^{h} r^G(q), \qquad M(h) = \frac{K_Y(h)}{K_G(h)}.$$

Table D.1 reports the estimation setup. Table D.2 reports eighth-year coefficient estimates and normalised response ratios for the linear benchmark and the four individual transmission channels.

Table D.3 summarises pointwise p values by channel and outcome over horizons $h = 0, \ldots, 8$. Table D.4 reports the eighth-year sample diagnostics and Poland profile values. Table D.5 shows how the retained response paths translate into cumulative $K_Y(8)$, $K_G(8)$, and $K_Y(8)/K_G(8)$ values.

Appendix D reports pointwise, horizon-level diagnostics. Because local projections estimate separate coefficients by horizon, the standard errors and p values in these tables describe each displayed coefficient at that horizon. They are not simultaneous confidence bands, path-level tests, or no-effect tests for the full response sequence (Jorda, 2005, 2007; Montiel Olea and Plagborg-Moller, 2021; Inoue, Jorda and Kuersteiner, 2024; Barnichon and Brownlees, 2019; Li, Plagborg-Moller and Wolf, 2024). Related fiscal and public-investment multiplier papers likewise report response paths with confidence bands, p values, or horizon-specific significance indicators when precision varies over the horizon (Deleidi, Iafrate and Levrero, 2020; Ciaffi, Deleidi and Di Domenico, 2024; Saccone, 2022).

Table D.1. Estimation setup for the 2025 mixed-window specification.

State-variable channel	Horizon-zero shock years	Countries	Controls and fixed effects	Use of the 2025 observation
Linear EU27 benchmark	2004-2025; the endpoint moves back with the response horizon	27 EU countries	Public-consumption shock, lagged investment growth, lagged public-consumption growth, lagged output growth, lagged short-term rate, country fixed effects, year fixed effects	2025 is used where the outcome, shock and controls are observed; this benchmark has no state-variable requirement
State variables not using investment import content	2004-2025; the endpoint moves back with the response horizon	27 EU countries	Public-consumption shock, lagged investment growth, lagged public-consumption growth, lagged output growth, lagged short-term rate, country fixed effects, year fixed effects	2025 is used where Eurostat provides the required macroeconomic and financial-account data; Ireland remains in rows where the needed fields exist
State variables using investment import content	2004-2023 at horizon zero; the endpoint moves back with the response horizon	27 EU countries	The same controls and fixed effects, plus the state variable and its interaction with the public-investment shock	The trade state variable uses the official OECD TiVA source window through 2022

State-variable channel	Horizon-zero shock years	Countries	Controls and fixed effects	Use of the 2025 observation
Polish 2025 profile used for state-specific evaluation	Profile year 2025 for Eurostat state variables; official 2022 TiVA value for the trade profile	Poland profile against the EU panel standardisation	The profile is applied to the estimated response equations; it is not a separate regression sample	The mixed profile lets financial-account and macroeconomic variables use 2025 and uses the official 2022 TiVA trade profile

Table D.2. Eighth-year coefficient estimates and normalised response ratios.

Channel evaluation	Outcome	h	Public-investment coefficient	Coefficient p value	Normalised response ratio	Ratio p value	Observations	Countries	Years
EU27 linear benchmark	Output	8	0.009528 (0.005564)	0.086814	0.225432 (0.127300)	0.076584	378	27	2004-2017
EU27 linear benchmark	Public investment spending	8	0.002815 (0.001690)	0.095893	0.066592 (0.040694)	0.101755	378	27	2004-2017
Investment import content	Output	8	0.013313 (0.003932)	0.000708	0.314667 (0.085817)	0.000246	378	27	2004-2017
Investment import content	Public investment spending	8	0.003045 (0.001624)	0.060773	0.071969 (0.039403)	0.067781	378	27	2004-2017
Household liquidity position	Output	8	0.013000 (0.003780)	0.000584	0.301412 (0.083828)	0.000324	378	27	2004-2017
Household liquidity position	Public investment spending	8	0.002237 (0.001687)	0.184739	0.051868 (0.039422)	0.188267	378	27	2004-2017
Public debt	Output	8	0.008337 (0.005907)	0.158108	0.197160 (0.138106)	0.153409	378	27	2004-2017

Channel evaluation	Outcome	h	Public-investment coefficient	Coefficient p value	Normalised response ratio	Ratio p value	Observations	Countries	Years
Public debt	Public investment spending	8	0.002680 (0.001352)	0.047404	0.063378 (0.031859)	0.046668	378	27	2004-2017
Real PPP income	Output	8	0.002272 (0.006167)	0.712511	0.055751 (0.150230)	0.710561	378	27	2004-2017
Real PPP income	Public investment spending	8	0.005604 (0.002284)	0.014162	0.137493 (0.060154)	0.022272	378	27	2004-2017

Table D.3. Pointwise p-value summary by channel and outcome.

Channel evaluation	Outcome	Horizons	Coefficient p values above 0.10	Ratio p values above 0.10	Minimum observations	Maximum observations	Countries	First year	Last year
EU27 linear benchmark	Output	9	6	6	378	583	27	2004	2025
EU27 linear benchmark	Public investment spending	9	5	6	378	583	27	2004	2025
Household liquidity position	Output	9	6	6	378	583	27	2004	2025
Household liquidity position	Public investment spending	9	6	6	378	583	27	2004	2025
Investment import content	Output	9	4	4	378	531	27	2004	2023

Channel evaluation	Outcome	Horizons	Coefficient p values above 0.10	Ratio p values above 0.10	Minimum observations	Maximum observations	Countries	First year	Last year
Investment import content	Public investment spending	9	5	5	378	531	27	2004	2023
Public debt	Output	9	7	7	378	583	27	2004	2025
Public debt	Public investment spending	9	5	5	378	583	27	2004	2025
Real PPP income	Output	9	8	8	378	583	27	2004	2025
Real PPP income	Public investment spending	9	6	6	378	583	27	2004	2025

Table D.4. Eighth-year sample diagnostics and Polish profile values.

Channel evaluation	Poland standardised state value	Observations	Countries	Shock years
EU27 linear benchmark	NA	378	27	2004-2017
Investment import content	-0.161	378	27	2004-2017
Household liquidity position	0.576	378	27	2004-2017
Public debt	-0.081	378	27	2004-2017
Real PPP income	0.074	378	27	2004-2017

Table D.5. From regression responses to cumulative output and spending paths at horizon eight.

Path	$K_Y(8)$	$K_G(8)$	$K_Y(8)/K_G(8)$	Role
EU27 benchmark	2.225433	0.776057	2.867616	EU27 reference
Investment import content	1.797840	0.726559	2.474458	Retained Polish channel evaluation
Household liquidity position	2.235754	0.767188	2.914218	Retained Polish channel evaluation
Public debt	1.781344	0.752662	2.366726	Retained Polish channel evaluation
Equal weight average	1.938312	0.748803	2.588548	Equal weight average of retained Polish channel evaluations

Notes: Parentheses in Table D.2 report the standard error for the displayed coefficient or response ratio; the pointwise, horizon-level p value is reported in its own column. Non-TiVA specifications use 2025 observations where the required Eurostat variables are available, including Ireland where the needed fields exist. Specifications using investment import content use the official OECD TiVA source window through 2022. The cumulative response translation uses the same horizon-specific response estimates to construct the output and spending paths used in the debt application.